Understanding Everything

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Preface

1 "Everything" and "understanding"

This book is for those who want to understand everything. By "everything" I mean mathematics, physics, chemistry, biology, psychology, astronomy, meteorology, engineering, computer science, cats and dogs and art and dreams and every other object a mind can apprehend. Yes, absolutely everything: the universe, big and small, mundane and arcane.

I must immediately clarify that such complete understanding is probably impossible,* and it is not the purpose of this book to provide it. Rather, my intended readership is those who knowingly pursue this impossible end, and I wrote this book to assist them as much as I can.

So, what does it mean to "understand" something?† First, you need to know its behaviour. More specifically, you must be able to predict, as precisely and accurately as physically possible, how it will interact with the rest of reality in any given circumstance. If the difficulty of satisfying this first criterion is not obvious, I recommend you try it with the weather and record your accuracy.

Second, you have to know why it has that behaviour. To ask "why" is to be curious, and to keep asking is to demonstrate greater curiosity. A girl asks her father why rain falls. He explains things about evaporation, convection, condensation and gravity. She may continue asking "why" until her father has explained the behaviour of subatomic particles, at which point one more question will force him to admit he has no idea. In writing this book, my aspiration has been to satisfy this child who unrelentingly asks "why?"

2 How to proceed

Suppose you're a medical student, and you want to understand the human body. This is an extremely ambitious goal which I doubt any physician has ever accomplished. The human body is complex to a literally unimaginable degree. It is composed of organs: skin, a heart, lungs, and so on. Each organ is composed of tissues, and these, in turn, of cells. Each cell is a complex system made of organelles, which are built from proteins. Proteins are made of molecules, which are composed of atoms. Perhaps finally,‡ each atom is composed of subatomic particles. As we view the body at finer and finer resolution, its components multiply in number and interact with brainshattering intricacy.

How can you proceed? First, you will automatically become familiar with the body by being one yourself. You may also observe the body in various situations and see how it reacts. Perhaps you notice that when someone regularly tries to lift slightly heavier weights,

*

For numerous reasons including the following:

1. Certain phenomena may be fundamentally unpredictable, or otherwise unknowable.
2. It may be that, even under optimal conditions, understanding everything would require more time than we have left before the heat death of the universe.
3. It might be that no mind, regardless of augmentation, is capable of containing a simultaneous understanding of all things.
4. My definition of "understanding," given below, might be so strong as to be unattainable, or so weak as to be unsatisfying.
5. It may be that human motivation is insufficient for the task: that is, we might simply get bored.

†

Of course, understanding is a spectrum. Here I provide a particular standard that we can strive for within this book. I do not claim that this is the "correct" definition of understanding, nor that it is even possible to attain, nor that my methods are well-suited to attain it. It is simply a clarification of the goal of this book.

‡

More on this "perhaps finally" below.

eats and sleeps enough, and takes enough time to recover, they grow bigger and stronger. Why? Well, you know from your studies that generally the body adapts to its circumstances as long as it has the resources to do so. But why? To answer, you need to look deeper: at organs, cells and organelles.

This is a general pattern. To understand something, it is necessary but insufficient to observe it as a whole. You must also break it down into parts and understand the parts. Therefore, consider the following scheme for understanding any thing.

1. Familiarize yourself with its behaviour as a whole.
2. Decompose it into parts.*
3. Understand each part (following this same procedure).
4. Figure out how the parts interact to produce the behaviour of the whole.

According to step 3, to understand something you must understand its parts. To understand each part, you must understand *its* parts in turn and so on until you reach indivisibles. It is possible that there are no indivisibles, in which case this process never ends. However, humanity has so far been unable to decompose subatomic particles into parts. Therefore, if you wish to carry the above procedure deeper and smaller than these, your activity will change sharply at that point from study to research. This book limits itself to facilitate the former. Hence, for our purposes, subatomic particles may be considered indivisible. And indeed, as far as we know, subatomic particles are the fundamental constituents of absolutely all physical objects. So to understand anything you will ultimately need to understand subatomic particles.

Therefore this book falls partially under the purview of physics. And traditionally, physics education more or less follows our 4-step procedure above. A physics student first learns about things very near the top of the complexity hierarchy: how baseballs move when thrown through the atmosphere of Earth, and so on. Then gradually one learns more and more general truths about smaller and smaller things: about momentum and gravity, then special relativity and electromagnetism, and eventually quantum field theory and the Standard Model of particle physics.

Unfortunately, the experience of a student undergoing such an education is often one of increasing confusion and difficulty. It begins with familiar objects and simple situations, and ends with quantum mysteries that no one understands. As the program progresses, teachers retread the same topics over and over again, each time oversimplifying slightly less. In the mean time, the student must memorize increasingly strange equations for which no explanation can be given until they take a more advanced class. Mathematics is generally avoided as much as possible, ill-preparing students for the increasingly abstract mathematics necessary for advanced physics.†

We will take the opposite approach, an inversion of the 4-step process above. You will first thoroughly study mathematics in its own right, fully preparing you for physics and much else besides. Next come the most general truths about subatomic particles, explained properly in the language of mathematics now familiar to you. Finally, we will use this understanding to explain atoms, then molecules and increasingly composite systems until we reach things like baseballs and human bodies.

*

Exactly how we do this is somewhat arbitrary. Should the body be broken up into organs, or first into organ systems? And if we choose organ systems, in which system should we place the kidneys? We can break things down however we want, and we should use this freedom wisely.

As for the question of scale (e.g., organ systems vs. organs), I propose that we always make the smallest possible step. That is, we should not jump from organs to atoms: there are too many atoms in an organ to make sense of. Instead, we'll use stepping stones in-between: tissues, cells, organelles, proteins and molecules. That way, at any given stage we will only have to think about a manageable number of parts. Therefore, I would break the human body down first into organ systems, not directly into organs.

To answer the question of what goes where, I propose we always choose parts that make sense as unified wholes in and of themselves. This is because, to understand each of those parts (see step 3), we will need to break them down into subparts and then recombine those subparts to form a whole (see step 4), and this will be easier if there is a sensible whole to form. Therefore I would place the kidneys in the urinary system, because of how they work together with certain other organs (the bladder, urethra and ureters) to perform a coherent function as a whole (to clean the blood).

†

I make these criticisms only to motivate my approach, not to disparage those who choose another.

The motivation for the traditional approach is to start simple, and my reasoning is the same: subatomic particles are the simplest things in existence, because any other object must be composed of them and is therefore more complex. They are admittedly less familiar and therefore more intimidating to some, but to my intended readership this will be no obstacle. We will discuss each subject only once, properly and rigorously the first time.*

3 Tips for the reader

1. Read the sidenotes and footnotes. They are not optional, and lie in the margins only so that on a second reading they don't interfere with the main text. Some of the most important sentences in this book are to be found there, and my explanations will assume you have read them.

   If you ever cannot find a sidenote, look on the next page. Sometimes there are so many that they spill over.

2. This book is intended to be read in strict sequence, but feel free to look ahead† (and certainly behind).

3. Don't intentionally memorize. If it's worth remembering, you'll soon remember it anyway.‡

4. Every concept in this book is meant to be understood before you move on. If it doesn't need to be understood, I don't bother writing about it. Never be content to not entirely understand. Misunderstandings will accumulate and you will find yourself lost. Everything is supposed to make sense.

5. Do a lot of thinking. Some books do that for you, leaving little unsaid. This is not one of them. I expect you to think carefully about what you are reading. Make hypotheses and test them. Generalize. Think of examples. Learn how you learn, learn how to understand, and do it.

6. Don't worry about moving slowly, especially not relative to others. You are taking the time to truly understand, and usually those who move faster don't care so much about understanding things, or are only pretending to understand, or do not know how understanding something feels. If you dedicate yourself to this book, you will be continually surprised at how many highly educated people do not understand even the basics of their subject.§

7. This book contains many exercises, which are embedded in the text by indicators like this: Do Exercise 3.1. Do every single one:¶ I will not waste your time with busywork. Indeed, almost all of the important mathematical results in this book are to be found in the exercises. You will discover these truths for yourself. I will merely guide your investigation.

   When you reach an indicator, flip to the end of the section and do the indicated exercises *before reading on*. Do not keep going and come back later. Often there will be spoilers if you do.

*

Exceptions to this rule are necessary. For example, logic (which is a language) is best described in logic. I would love to have it introduce itself to you in its own words, but you wouldn't know what it was saying. Therefore, at first I will speak for logic, and later we'll bring it back to speak for itself and correct my imprecisions.

†

Although you may not understand what you find there, and shouldn't let this discourage you.

‡

More generally, if you think something is necessary, try not doing it and see if anything bad happens.

§

I do not mean this as an insult. I myself am a computer scientist who does not (yet) understand computers.

My intention is purely to provide solace to the student who works carefully and watches as others seem to pull ahead. They are not moving faster; they are moving in a different direction.

¶

Yes, including that one.

*

This preface is not considered a chapter.

†

This is, of course, rather risky. Allow me to enumerate the dangers, and how I seek to mitigate them:

1. If the answers cannot be trusted, why look at them?

   First of all, the vast majority of the answers are correct, and it is vital you read them: to learn how to think about mathematics, how to solve problems, and how to reason in general.

   Second, although you cannot trust any single solution in isolation, you can notice patterns of reasoning across several solutions and deduce which solution involves reasoning that doesn't fit those patterns.

   Third, having detected an incorrect solution, you can thereby trust the others more fully. I will not introduce an incorrect solution until you are prepared to detect it.

2. If you trust an incorrect solution, your intuition might be poisoned.

   To mitigate this, I have placed a "List of incorrect solutions" at the back of the book. After you've worked through a chapter and tried to suss out its incorrect solutions yourself, you should *always* consult this list and re-examine any solutions that you were mistaken about.

‡

From the Greek ἔτυμος, meaning *true* or *real*.[a]

[a] I recommend you get used to self-reference—such as the etymology of "etymology" (or notes inside notes (or parentheticals inside parentheticals))—early, before things get out of hand.

You are not expected to write your solutions to the exercises down somewhere. Of course, use paper if you need to, but I recommend not doing so unnecessarily. Finding solutions entirely within your mind makes you stronger and less dependent. I find it very pleasant (and immensely cognitively beneficial) to do this while walking outside.

Many exercises are given fully worked solutions at the back of the book. These take the place of worked examples in other textbooks: instead of showing you how to do something before you do it, I first give it to you as an exercise and then explain how to do it in the solution. It is imperative that you do all you can to solve each exercise before looking at its solution, no matter how unprepared, how aimless, you may feel. That being said, once you've completed each exercise, I do expect you to read the solution before moving on. Often, I will include important content in these solutions, and refer to them later.

I include these solutions to demonstrate useful thought processes. It is *not* so that you can check your work. Do not misunderstand me: checking your work is necessary for refining your problem-solving process. If your conclusions are not tested, you can end up anywhere. Rather, *because* of the importance of verifying your work, I want you to learn to do it yourself. Therefore, **in each chapter,* at least one of the solutions is deliberately incorrect.** If you accept them uncritically, you will be led astray.†

Furthermore, some exercises are ambiguous, open-ended, or subjective. Your task is to recognize them as such and handle them with nuance.

8. Use every resource at your disposal and do whatever is necessary. If you do not know a word (or don't know how to pronounce it), look it up. If you don't understand something, do whatever research or thought is necessary to figure it out. Skip around, reread things. If you notice a truth, use it.

Exercises

3.1 ⑤ Find where I keep the solutions and seek out the solution to this exercise.[a]

[a] Many exercises will have notes attached, like this one. They are meant to be read before attempting the exercise. While you're here, two more pieces of information:

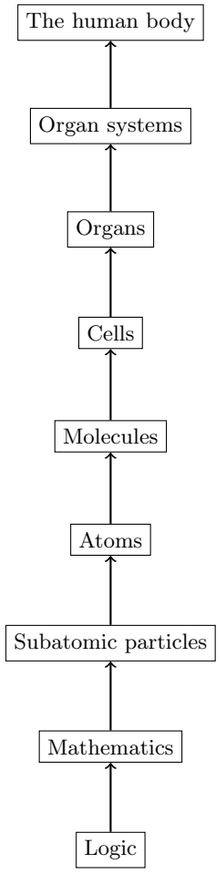
1. Exercises are numbered in the format "`[section].[exercise]`". Therefore, this is the first exercise in section 3.
2. The ⑤ at the start of this exercise indicates that it is given a solution at the back of the book. Exercises without solutions are instead marked by ◯.

4 To stand under

One more thing, before we begin. Consider the etymology‡ of the word "understand": to *stand under*. This is an old metaphor, and its sense isn't obvious. Yet if you have understood this preface, you will be able to see my interpretation with some thought. Do Exercise 4.1.

Exercises

4.1 ○ Who is standing under what, and in which sense?



Chapter 1

Mathematics and logic

1 What is mathematics?

I think of mathematics as a process of rigorously investigating very clear concepts.* A mathematician begins their work by clearly defining which things they want to investigate by asserting a set of basic assumptions, called *axioms*, about those things. The things under investigation are called *mathematical objects*, and are usually (but not always) chosen to represent some intuitive idea, such as the idea of a number. The axioms are then chosen to accurately capture the essence of that idea.

Example. If our mathematical objects represent numbers, our axioms should capture the essence of numbers.†

Axioms are written in the language of *logic*. Logic is simple, unambiguous, and well-suited for reasoning, and this allows us to rigorously deduce new mathematical facts, called *theorems*, from the axioms. Theorems are also written in logic, and so from our first theorems we can rigorously deduce further theorems, and so on ad infinitum. The collection of theorems so obtained constitutes a *theory* of the represented idea.‡ If our axioms are accurate and our deductions are valid, then the resulting theory§ will accurately describe the represented idea. We can therefore learn new information about an idea by capturing its essence with axioms and examining the resulting theory.

Examples. Here are some examples of mathematical theories.

1. Number theory, the theory of whole numbers
2. Complex analysis, the theory of numbers in general
3. Geometry, the theory of space
4. Topology, the theory of surfaces
5. Set theory, the theory of collections

As we go, we also define new *mathematical concepts*¶ to investigate.

Example. While investigating numbers, we might define the *even numbers* as those divisible by two.

*

You should read the preface if you haven't.

†

It's important to distinguish between the mathematical objects defined by the axioms and the idea they are supposed to represent. This is because ideas are vague, so there are many possible *axiomatizations*[a] for any given idea.

Two axiomatizations of the same idea might define entirely different mathematical objects. One axiomatization might be more accurate than the other, or they might be two mutually enlightening angles from which to view one idea.

[a] That is, choices of axioms.

‡

Note that I did not say *the* theory, since different axiomatizations can result in a variety of theories of the same idea.

§

Here I said *the* resulting theory: from any given axiomatization, there is a single collection of theorems that logically follow.[a]

[a] Which theorems we *find* is another question: mathematical theories are so vast that we only scratch their surfaces.

¶

Mathematical concepts[a] are what we investigate in mathematics, and are to be distinguished from the intuitive ideas that we indirectly learn about by doing so. They include the original objects of a theory but also newly defined concepts.

[a] The reason I do not call these simply mathematical *objects* is that often the defined concepts are not very object-like.

Such new concepts must be unambiguously (that is, logically) defined in terms of concepts that are already meaningful within the theory: either our original objects or previously defined concepts.

Example. The even numbers are defined in terms of:

1. Numbers, which are the basic objects of the theory.
2. The number 2, which is one such number.
3. Divisibility, which is a previously defined concept.

Like the original objects, defined concepts are often chosen to represent an intuitive idea. However, it is also common to start without any such interpretation in mind. In that case, our first intuitions* about the new concept come from our existing understanding of the concepts in whose terms it was defined. Then, because its definition was stated in logic, we can deduce theorems about it and learn more. In this way, mathematicians construct and investigate a tower of genuinely novel concepts.

This is fortunate, because the concepts of natural language are incapable of expressing the behaviour of subatomic particles. The grammar and words of English are designed to facilitate communication about everyday life, and are ill-equipped for the task of describing such unfamiliar objects as quarks. But, having thoroughly investigated the mathematical edifice, we will be able to draw upon a comprehensive library of concepts—each thoroughly understood both in its own right and in relation to the others. With these, we can construct precise models of reality immediately rich with meaning and consequence.

Some mathematical concepts are more useful to study than others. In particular, the concept of a *set*, interpreted as a collection of things of any kind, is sufficient to define almost all the rest of mathematics in its terms. In this way, sets are analogous to subatomic particles: having understood sets, we can understand practically everything in mathematics, and certainly everything we will concern ourselves with in this book.

That is why the first Part of this book, beginning with Chapter 2, is devoted to sets. Having acquired a thorough understanding of sets, you will be able to enter every other subfield of mathematics easily and with a sense of familiarity. But to understand the axioms of set theory,† you must first learn to read and write logic.

*

By your "intuition," I mean your understanding, and what you think or feel without proof.[a] One's intuition is honed by experience: an untrained intuition is usually wrong, but a trained one is often right.

A well-developed intuition is essential for a mathematician, because before you prove something you must decide what to prove, and there are infinitely many propositions you might select and almost all of them are false. It is intuition that guides this decision and many others.

[a] A *proof* of a theorem is a rigorous explanation of why it is true. We will discuss proofs at the end of this chapter.

†

By "set theory," I mean Zermelo-Fraenkel set theory with the axiom of choice, which is the theory resulting from a particular axiomatization of the idea of a set, devised by Ernst Zermelo and Abraham Fraenkel.

‡

By "pizza is delicious," one might mean that they find it delicious, or that most people do, and *these* claims are propositions. You can see why mathematicians prefer to use a less ambiguous language.

2 Propositions and propositional logic

Logic is an easy language to learn, compared to natural languages like English. Its words are simple and few. In English, words combine to form sentences that express claims, questions, commands, exclamations, etc. In contrast, logic can only express *propositions*.

A *proposition* is a claim of objective fact: a description of reality which the speaker claims to be accurate.

Examples.

1. "I love pizza" is a proposition, because love is a biological phenomenon that either occurs or does not.
2. "Pizza is delicious" is not a proposition, because there is no part of reality that could be tested to settle the matter.‡ Do Exercise 2.1.

In logic, we distinguish two kinds of proposition:

1. *Atomic propositions*,* and
2. *Composite propositions*, which are composed by applying *logical operators* to one or more existing atomic propositions.

Perhaps surprisingly, we will first discuss how to combine atomic propositions via the logical operators, and only later discuss how to form those atomic propositions in the first place.† To do so, we temporarily ignore the inner details of atomic propositions, and distinguish them only by whether they are true or false. That is, we consider all true statements to be equivalent, and all false statements to be equivalent. This portion of logic is called *propositional logic*, because within it we deal only with propositions, as opposed to the other logical ideas needed to expose their details.

*

So named because they constitute more complex propositions, like atoms constitute molecules.

†

Partly this is because the atomic propositions depend on the mathematical theory we are interested in, while the logical operators are more generally applicable. Secondly, many atomic propositions *involve* logical operators, as you will soon see.

‡

There is much more to be said about truth and falsehood, but much of it involves concepts which we will have to build up to at length. Therefore, we will defer these discussions until later.

§

Why that name?

¶

So named because it is tacitly assumed that any alternative must be *between* truth and falsehood, whatever that may mean.

Exercises

2.1 ⓢ Determine which of the following expressions are propositions.

1. Who are you?
2. Water is wet.
3. Close the door.
4. $2 + 2 = 1000$.
5. Proposition.
6. Murder is wrong.
7. This sentence is not true.[a]

[a] To be clear, "this sentence" refers to itself.

3 Truth values and propositional variables

In mathematics, we usually assume that propositions are either true or false: they cannot be both, nor neither. For a proposition to be *true* means that its description of reality is perfectly accurate: that is, the proposition portrays some part of reality as being a certain way, and that part of reality is in fact that way. For a proposition to be *false* means that its description of reality is inaccurate.‡

The first assumption, that a proposition cannot be both true and false, is called the *Law of Noncontradiction*.§ This seems straightforwardly correct: reality cannot both be and not be the way a sentence portrays it; either the description is accurate or it isn't. Intuitively, the Law of Noncontradiction is obvious, and *empirically* (that is, in human experience) it has never failed. Indeed, it is impossible to imagine how it could.

The second assumption, that there are no other options besides truth and falsehood, is called the *Law of the Excluded Middle*.¶ Again, this seems pretty self-evident: what would it mean for a clear, meaningful description of reality to be neither accurate nor inaccurate? You might imagine it could be *partially* accurate, but in such cases we simply call it inaccurate.

For example, if I say that "the sky is always red and the sun is hot," my overall claim is simply false: reading this claim, we imagine a reality that is not our own.

Therefore, within this book, we will make both of these assumptions. However, I will take a moment to address one important counterargument (although we have already rebutted it by accident). This argument describes a strange situation called the *Liar Paradox*, which is created by the Liar Sentence: "This sentence is not true."

Under the laws of Noncontradiction and of the Excluded Middle, we assume that every proposition is either true or false but not both. Therefore (so the argument goes), the Liar Sentence is either true or false (but not both).

But suppose it is true. Then its self-description must be accurate: that is, it must not be true. So, if it is true, it must also be false (for that is the only remaining option). But if it is false, then its self-description must be inaccurate: that is, it must be true! Hence, in *either case*, the Liar Sentence must be both true and false. Do Exercise 3.1.

Having been convinced by this argument, many logicians and mathematicians have explored *non-classical logics*: that is, logical systems in which one or both of the "Laws" are not assumed. For example, there are "Many-Valued Logics," in which further options (or *values*) are allowed in addition to truth and falsehood, such as "indeterminate", "both", or "neither". Some have even proposed an infinite number of values in-between truth and falsehood, to represent shades of accuracy.

Others have concluded for various reasons that the Liar Sentence is simply not a proposition, and that therefore the paradox doesn't reveal a problem with our logic but a confusion about our language. This, indeed, is the position taken in this book. Therefore, every proposition has to make a "choice" between these two possibilities: truth or falsehood.* Then, whether a given statement is true or false is called its *truth value*.

Example. The truth value of "the Earth orbits the Sun" is that it is true.

For brevity, we give names to the two truth values: $\top$ for Truth, and $\bot$ for Falsehood. Then, to translate an English proposition into propositional logic, we just write its truth value. That is, all true propositions are denoted $\top$, and all false ones are denoted $\bot$.

Examples. Here's how we would translate some English propositions into propositional logic.

1. For "the sky contains air," we will just write $\top$, because it is true.
2. Similarly, for "cats are reptiles" we will write $\bot$. Do Exercise 3.2.

Sometimes we won't know the truth value of some statement, such as "subatomic particles are indivisible." To translate it into propositional logic, we cannot confidently write either $\top$ or $\bot$. Therefore, we write some letter as a placeholder, called a *propositional variable*.† By convention we tend to use p,‡ q, r, and so on up the alphabet as necessary. Therefore we should translate our statement about subatomic particles as p.

* Of course, generally this choice is made by reality, and the proposition has no say.[a]

[a] There are exceptions, which I will explain later in this chapter.

† A *variable* is a letter that represents some unspecified mathematical object. We use the term "variable" because there is no fixed object that the letter represents: it can *vary*. We call these ones specifically *propositional* variables because the objects that they refer to are propositions.[a]

[a] Strictly speaking, the propositions we're dealing with are not mathematical objects, because we have not defined them with axioms. Much later, we will do so.

‡ For "proposition."

Exercises

3.1 ⓢ Is the Liar Sentence true, false, both, or neither?

3.2 ◯ Determine the truth value of each proposition from Exercise 2.1. If you don't know the truth value of a proposition, skip it.

4 Composite propositions and logical operators

Consider the statement "octopuses have three hearts and live for decades." This is an example of a (false) composite proposition: two propositions conjoined by the logical operator "and". In this book, we concern ourselves with only five logical operators, each represented by its own symbol. They are as follows:

1. The **denial** of p, denoted "$\neg p$" and pronounced "it is not the case that p," asserts that p is false. This assertion is false when p is true. Do Exercises 4.1–4.2.

2. The **conjunction**[*] of p and q, denoted "$p \wedge q$" and pronounced "p and q," asserts that both p and q are true. Therefore it is false when one or both of p and q are false. Do Exercises 4.3–4.4.

3. The **disjunction**[†] of p and q, denoted "$p \vee q$" and pronounced "p or q," asserts that at least one of p and q are true.[‡] It is false only when both p and q are false. Do Exercises 4.6–4.7.

4. The **conditional** from p to q, denoted "$p \implies q$" and pronounced "if p then q," asserts that it is not the case that its premise (p) is true and its conclusion (q) is false. That is, $p \implies q$ is synonymous with $\neg(p \wedge \neg q)$. The concept of the conditional is difficult for most students the first time they encounter it. Be gentle with yourself and take your time. Do Exercises 4.8–4.16.

5. The **biconditional** between p and q, denoted "$p \iff q$" and pronounced "p if and only if q," asserts that p and q have the same truth value—that is, that they are both true or both false.

   Examples. Here are a few examples of using the biconditional to (accurately) state the truth value of a composite proposition.

   1. $\neg\top \iff \bot$.[§]
   2. $\neg\neg\top \iff \top$.
   3. $\top \wedge \top \iff \top$.
   4. $\top \wedge \bot \iff \bot$.
   5. $(\bot \implies \top)$[¶] $\iff \top$. Do Exercises 4.17–4.23.

Examples. Here are some translations of composite propositions from English to propositional logic.

1. "The sun is not hot" becomes "$\neg\top$."
2. "Octopuses have three hearts and live for decades" becomes "$\top \wedge \bot$."
3. "Two is less than three but not less than two" becomes "$\top \wedge \neg\bot$."[‖]
4. "If $2 + 2 = 4$, then pigs can fly" becomes "$\top \implies \bot$."

*
That is, the "joining-together."

†
That is, the "pulling-apart." I admit this terminology makes little sense to me.

‡
In English there are two kinds of "or."

1. *Inclusive*: "A or B, or both."
2. *Exclusive*: "A or B, but not both."

In "Do your job or you'll be fired," one hopes that these options are mutually exclusive. But in "Students must take MATH 3141 or CSCI 1024," it is acceptable to take both. Do Exercise 4.5.
In mathematics, "or" is always inclusive.

§
This biconditional claims that $\neg\top$ and $\bot$ have the same truth value, which amounts to saying that $\neg\top$ is false.

¶
We use parentheses for clarity.

‖
"But" and "and" both translate to "$\wedge$", though the former implies contrast and the latter does not. When translating into logic, ignore logically irrelevant nuances.

5. "He can neither read nor write" becomes "$\neg p \wedge \neg q$."*
6. "We can go to the movies, or not" becomes "$p \vee \neg p$."†
7. "n is even if and only if it is divisible by 2" becomes "$p \iff q$."‡

*

Here we use variables because we don't know if this guy can read or write.

†

Don't translate this as $p \vee \neg q$. It is very important that the two propositions are the same. Do Exercises 4.24–4.25.

‡

We will find biconditionals very useful for stating definitions.

Exercises

4.1 ⓢ If p is false, what must be the truth value of $\neg p$?

4.2 ⓢ What is the truth value of $\neg\neg\bot$?

4.3 ⓢ If p is true, can $p \wedge q$ be false?

4.4 ⓢ If p is false, can $p \wedge q$ be true?

4.5 ⓢ Which "or" do mathematicians assume stands between truth and falsehood?

4.6 ⓢ If p is true, can $p \vee q$ be false?

4.7 ⓢ If p is false, can $p \vee q$ be true?

4.8 ⓢ Assuming that $p \implies q$ is false, deduce the truth values of p and q.

4.9 ⓢ If p is true, and $p \implies q$ is also true, deduce the truth value of q.

4.10 ⓢ If q is false, and $p \implies q$ is true, deduce the truth value of p.

4.11 ⓢ Can a conditional be false while its premise is false?

4.12 ⓢ Can a conditional be false while its conclusion is true?

4.13 ⓢ In "$p \implies q$", why do we call q the "conclusion"?

4.14 ⓢ Why do we call "$p \implies q$" a "conditional"?

4.15 ⓢ What is the relation between a logical conditional and an English *if... then* statement? (Consider the example "The sun is hot $\implies$ these words are written on a page.")

4.16 ⓢ What does it mean if $p \implies q$ and $q \implies p$? (Notice I did not say "...if $p \implies q$ and $q \implies p$ <u>are true</u>," because that would be redundant: to claim "p" is to claim that "p is true." For example, if I tell someone that my name is Cuyler, it is implied that I am saying that the statement "my name is Cuyler" is true.)

4.17 ⓢ Why is this operator called the "biconditional"?

4.18 ⓢ Why do we pronounce this operator as "if and only if"?

4.19 ⓢ When is $p \iff q$ false?

4.20 ⓢ If p and $p \iff q$, deduce the truth value of q.

4.21 ◯ If $\neg p$ and $p \iff q$, deduce the truth value of q.

4.22 ◯ If q and $p \iff q$, deduce the truth value of p.

4.23 ◯ If $\neg q$ and $p \iff q$, deduce the truth value of p.

4.24 ⓢ Can $p \vee \neg p$ be false?

4.25 ⓢ Can $p \vee \neg q$ be false?

5 Truth tables

Truth tables can help us understand the logical operators better. Here is an example that illustrates the behaviour of conjunction:

p	q	$p \wedge q$
$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\bot$
$\top$	$\bot$	$\bot$
$\top$	$\top$	$\top$

Each row lists one of the four possible combinations of truth values for p and q, and the corresponding truth value of their conjunction. Do Exercises 5.1–5.5.

We can also build truth tables for more complex formulas,* like so:

p	q	$p \implies q$	$q \implies p$†	$(p \implies q) \wedge (q \implies p)$
$\bot$	$\bot$	$\top$	$\top$	$\top$
$\bot$	$\top$	$\top$	$\bot$	$\bot$
$\top$	$\bot$	$\bot$	$\top$	$\bot$
$\top$	$\top$	$\top$	$\top$	$\top$

Do Exercise 5.6.

Exercises

5.1 ⓢ Why are there four combinations of truth values for the conjunction table?

5.2 ⓢ Construct the truth table for disjunction.

5.3 ⓢ Construct the truth table for denial.

5.4 ⓢ Construct the truth table for the conditional.[a]

5.5 ⓢ Construct the truth table for the biconditional.

5.6 ⓢ Compare this with your biconditional table.

[a] You can just add a column to your disjunction table, if you like. Do you see why? Could you have built your disjunction table by adding a column to your denial table?

*

A *formula* is something like "$p \vee q$": it is not itself a proposition, but becomes a proposition when its variables are replaced by truth values.

†

Here the goal is to fill the right-hand column, and these two intermediary columns are not strictly necessary.

However, they can be helpful for beginners. That is, you can break down a complex formula into parts, fill out the columns for those parts, and then use them to fill out the column for the original formula.

I would not include them in my own work and you may omit them whenever you want.

6 Logical equivalences

When two formulas, labeled F and G, have the same truth value for every combination of truth values of their variables, we call them *logically equivalent* and write "$F \equiv G$."* If you want to be able to see this at a glance, you can list your combinations in the same order in each table; or even better, put the columns for the two formulas in the same table. Then, if those columns are identical, the two formulas are logically equivalent; and if they differ, the formulas are not logically equivalent.

Examples. Here are two logical equivalences we already hinted at.

1. When I introduced the conditional, I said that $p \implies q$ was "synonymous" with $\neg(p \wedge \neg q)$. What I really meant was that $(p \implies q) \equiv \neg(p \wedge \neg q)$. To see this, let's make a truth table containing both formulas.

p	q	$p \implies q$	$\neg(p \wedge \neg q)$
$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\bot$
$\top$	$\top$	$\top$	$\top$

   Comparing the column for $p \implies q$ and the column for $\neg(p \wedge \neg q)$, we see that they are the same. Therefore, the two formulas are logically equivalent.

2. In Exercise 5.6, we found that the columns for $p \iff q$ and for $(p \implies q) \wedge (q \implies p)$ are identical. Therefore, $(p \iff q) \equiv [(p \implies q) \wedge (q \implies p)]$.

Logical equivalences are very useful in mathematics, primarily for two purposes:

1. When we first establish one, we can think about it and thereby improve our logical intuition.

2. We can use them to deduce new logical equivalences, and new theorems.

The second usage demands explanation.† Suppose we know that two formulas A and B are logically equivalent: $A \equiv B$. In other words, A and B always have the same truth value. Then we can derive a new equivalence by consistently replacing the variables in that equivalence with formulas. The resulting equivalence is called a "substitution instance" of the original.‡

Example. We know that $(p \implies q) \equiv \neg(p \wedge \neg q)$. For this discussion, we can think of A as denoting $p \implies q$ and B as denoting $\neg(p \wedge \neg q)$. If we replace p with $\neg p$ and q with $p \vee q$ throughout§ our equivalence above (that is, throughout A and throughout B), we obtain the following substitution instance:

$$(\neg p \implies (p \vee q)) \equiv \neg(\neg p \wedge \neg(p \vee q)).$$

Do Exercise 6.1.

*

This terminology is revealing. If two formulas have the same truth values in every circumstance, then for all logical and mathematical purposes they are synonymous: the same thing, written two different ways. Thus we say that they are *equivalent*, to evoke the idea that they are interchangeable (in the context of *logic*).

The symbol "≡" was chosen for its similarity with the "equals sign" in arithmetic, because logical equivalence and equality are themselves very similar.

For an immediate intuition as to why they are similar, note that both of them claim that two objects have the same *something*. A logical equivalence claims that two formulas always have the same truth value, and an equality claims that two numbers have the same size.

When we study set theory, we will find that logical equivalence and equality are two examples of what we call *equivalence relations*.

†

The process may seem complicated at first, but it's nearly identical to the usage of equality in arithmetic. The appearance of complexity comes from making that usage explicit. With experience, it will become automatic.

‡

This terminology evokes the idea that the original equivalence was a *more general* form, of which the new equivalence is a specific instance.

§

It is absolutely necessary that this replacement be *consistent* across all instances of each variable. That is, if the variable p occurs more than once, it must be replaced in each instance by the same formula. If you replace only one instance but not another, or replace each with different formulas, the resulting equivalence will be false.

Here is why this works. Before we made any replacements, the variables p and q each played a certain role in producing the overall truth values of A and B. That is, in every circumstance (i.e., every row in the truth table), each variable had some truth value that was then operated upon by various operators to determine the overall truth values of A and B. Our equivalence $A \equiv B$ tells us that, due to the particular way the operators *used* the values of p and q, every combination of those values resulted in A and B having the same truth value.

Having replaced p and q with formulas, those formulas now play the roles previously played by p and q. That is, the operators now use the truth value of the formula that replaced p (or q) the same way they originally used the value of p (or q). Therefore, the new versions of A and B, modified by these replacements, will always have the same truth value, for the same reason that A and B did.

In summary: if an equivalence is true, then its substitution instances are also true. This method of deduction will be called *variable expansion*.* This is already a very useful method of deduction, but we are not finished. I will now explain a second method, and then show you how to combine it with the first to form a more flexible and intuitive third.

The second method is as follows: if you have a formula F containing a subformula† S, and you know that S is equivalent to another formula S',‡ then, if you replace an instance§ of S with S' within F, this modified version of F (which we might call F') will be equivalent to the original. That is, $F \equiv F'$.

Example. Let F be the formula $(p \implies q) \wedge (q \implies p)$. Then $p \implies q$ is a subformula of F, and we know that $(p \implies q) \equiv \neg(p \wedge \neg q)$. Replacing $p \implies q$ with $\neg(p \wedge \neg q)$ within F, we obtain F': $\neg(p \wedge \neg q) \wedge (q \implies p)$. My claim is that, therefore:

$$[(p \implies q) \wedge (q \implies p)] \equiv [\neg(p \wedge \neg q) \wedge (q \implies p)].$$

Do Exercise 6.2.

We will call this method *subformula replacement*. It works for the same reason that the first method does: the original subformula S played some role in F, and now S' plays the same role within F'. I will leave the explanation at that, but please take as much time as you need to convince yourself.

Now we combine these two methods two produce a third. Suppose we have a formula F containing a subformula S. If we don't yet have an equivalence $S \equiv S'$ for some S' (and therefore cannot immediately apply the second method), but we do have an equivalence $G \equiv G'$ where S is a substitution instance of G,¶ then we can use the first method to create a substitution instance of $G \equiv G'$, which we might call $S \equiv S'$. *Then* we apply the second method, replacing S with S' within F to produce F' such that $F \equiv F'$.

Example. Let F be the formula $\neg(\neg p \implies \neg q)$. Then let S be the subformula $\neg p \implies \neg q$. We have not yet derived an equivalence involving this subformula S (and therefore cannot yet use subformula replacement), but we do know $(p \implies q) \equiv \neg(p \wedge \neg q)$, and S is a substitution instance of $p \implies q$. Specifically, S is obtained by replacing p with $\neg p$ and q with $\neg q$. Therefore we give the name G to the formula $p \implies q$, and the name G' to the formula $\neg(p \wedge \neg q)$, so that the above equivalence becomes $G \equiv G'$. Applying the appropriate substitutions to the equivalence $G \equiv G'$, we obtain:

$$(\neg p \implies \neg q) \equiv \neg(\neg p \wedge \neg\neg q).$$

*

Because by replacing a single variable with a larger formula we expand it.

†

That is, a formula within a formula. E.g., $p \wedge q$ is a subformula of $(p \wedge q) \implies p$. We also consider every formula to a be a subformula of itself, so that $p \wedge q$ is also a subformula of $p \wedge q$.

‡

The tick is pronounced "prime" so that S' is pronounced "S prime." In general, this notation indicates an alternate (or modified, etc.) version of something. In this particular case, S' denotes a formula different from, but logically equivalent to, S.

§

Sometimes a formula contains multiple instances of a subformula.

Example. $(p \wedge q) \iff \neg\neg(p \wedge q)$ contains two instances of $p \wedge q$.

¶

Previously I described one *logical equivalence* as a substitution instance of another. Here, in contrast, I describe one *formula* as a substitution instance of another. The meaning is directly analogous: a formula A is a substitution instance of a formula B if one can consistently replace the variables of B with formulas and thereby reproduce A.

The formula on the left side is simply S, and we will give the name S' to the one on the right. Now that we know $S \equiv S'$, we can use subformula replacement. Replacing S with S' throughout F, we obtain a new formula which we will call F':

$$\neg\neg(\neg p \implies \neg\neg q).$$

Because we obtained F' via subformula replacement, we conclude that $F \equiv F'$:

$$\neg(\neg p \implies \neg q) \equiv \neg\neg(\neg p \implies \neg\neg q).$$

Do Exercise 6.3.

In summary, if we have an equivalence $G \equiv G'$ and a formula F containing a subformula S that has the form of G,* then by replacing S with the corresponding S' that has the form of G', we produce a new formula F' equivalent to the original F. This third method will simply be called *substitution*. Its validity rests on the validity of the first and second methods from which it is composed.

So far, we have discussed methods of using logical equivalences to deduce new logical equivalences. But if equivalences were only useful for making more equivalences, they would not be very useful. Thankfully, we can also use them to deduce new mathematical *theorems* from old theorems. This is essentially due to the fact that a proposition can be a substitution instance of a formula.† If a proposition p has the form of a formula A, and we know that $F \equiv F'$, and also that p is true, then we can construct another proposition p' with the form of F' (using the same replacements used to derive p from F) and deduce that this p' is true.

Example. The proposition "If Socrates is a man, then he must die someday" has the form $p \implies q$: we obtain it by replacing p with "Socrates is a man" and q with "He must die someday" (and translating " $\implies$ " into an English *if... then* statement). Because $(p \implies q) \equiv \neg(p \wedge \neg q)$, we construct a new proposition of the form $\neg(p \wedge \neg q)$: "It is not the case that Socrates is a man yet will never die."‡ Finally, to the extent that you agree that all men die someday (due to the eventual heat death of the universe), we conclude that this new proposition is true.

Here is why this works. When we replace the variables in a formula with actual propositions, such as "Socrates is a man," those propositions have certain truth values (even if we aren't sure what they are) and therefore the formula's truth value is determined. That is, we narrow down its truth table to a single row. If we know that formula is equivalent to another, then they have the same truth value in *every* row, including this one. Therefore, if one is true, the other is true.

I will now ask you to *judge* a significant number of logical equivalences. Whenever, throughout this book, I ask you to "judge" a proposition, I mean for you to:

1. Figure out what it means, intuitively.
2. Decide whether you think it *should* be true, based on your intuition.
3. Determine its actual truth value.
4. When you have found the correct answer, turn it over in your mind until it is obvious. This will update your intuition.

* Instead of saying that S is a substitution instance of G, I will often say it has the *form* of G.

† That is, it can have the *form* of a formula. This is why we call them formulas: they represent a form that a proposition may have.

‡ A more natural paraphrase would be "It is *impossible* for Socrates to be a man and yet never die." This highlights the difference between an English *if... then* statement and the logical conditional.

This list of four steps is possibly the most valuable collection of words in this entire book, and I suggest you integrate them into your general problem-solving process. The first step alone will set you above the vast majority of thinkers: many problems become immensely easier when you take a moment to get clear about what is going on in the first place. The second and fourth steps are absolutely essential for internalizing and actually understanding much of anything in mathematics, and they will make problem-solving much easier.

This is your first major opportunity to reap the rewards of mathematical insight. **The entire purpose of studying mathematics is for these moments.** After putting in the hard work to determine what is true, you can then greedily absorb as much insight as you can from your findings. As you progress through mathematics, these rewards will grow ever more delicious.* Do Exercises 6.4–6.31.

*

You should start by using truth tables. Once you have a few equivalences, you can often use deduction instead.[a]

[a] That being said, sometimes it is easiest to just make the truth table. Usually it is a mistake to consider any technique universally superior to, or a replacement for, another.

Exercises

6.1 ⓢ Use a truth table to verify that this equivalence is true.

6.2 ⓢ Verify this with a truth table.

6.3 ◯ Verify this equivalence.

6.4 ⓢ Judge $p \wedge p \equiv p$.

6.5 ⓢ Judge $p \vee p \equiv p$.

6.6 ⓢ Judge $\neg\neg p \equiv p$.

6.7 ⓢ Judge $\neg\neg\neg p \equiv p$.

6.8 ⓢ Judge $\neg\neg\neg\neg p \equiv p$.

6.9 ⓢ Judge $p \wedge q \equiv q \wedge p$.

6.10 ⓢ Judge $\neg(p \wedge q) \equiv \neg p \wedge \neg q$.

6.11 ⓢ Judge $\neg(p \wedge q) \equiv \neg p \vee \neg q$

6.12 ⓢ Judge $p \vee q \equiv q \vee p$.

6.13 ⓢ Judge $\neg(p \vee q) \equiv \neg p \vee \neg q$.

6.14 ⓢ Judge $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

6.15 ⓢ Judge $p \wedge q \equiv \neg(\neg p \vee \neg q)$.

6.16 ⓢ Judge $p \vee q \equiv \neg(\neg p \wedge \neg q)$.

6.17 ⓢ Judge $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$.

6.18 ⓢ Judge $p \vee (q \vee r) \equiv (p \vee q) \vee r$.

6.19 ⓢ Judge $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.

*

As always, please take a moment to render this intuitively obvious for yourself.

6.20 ⓢ Judge $p \wedge (q \vee r) \equiv (p \wedge q) \vee r$.

6.21 ⓢ Judge $p \vee (q \wedge r) \equiv (p \vee q) \wedge r$.

6.22 ⓢ Judge $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

6.23 ⓢ Judge $p \implies q \equiv \neg p \vee q$.

6.24 ⓢ Judge $p \implies q \equiv q \implies p$.

6.25 ⓢ Judge $p \implies q \equiv \neg p \implies \neg q$.

6.26 ⓢ Judge $p \implies q \equiv \neg q \implies \neg p$.

6.27 ⓢ Judge $\neg(p \implies q) \equiv p \wedge \neg q$.

6.28 ⓢ Judge $\neg(p \implies q) \equiv \neg p \implies \neg q$.

6.29 ⓢ Judge $(p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$.

6.30 ⓢ Judge $p \iff q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$.

6.31 ⓢ What is the relationship between logical equivalence and the biconditional operator?

7 Tautologies and contradictions

Consider the following truth table.

p	$\neg p$	$p \vee \neg p$
$\bot$	$\top$	$\top$
$\top$	$\bot$	$\top$

That is, $p \vee \neg p$ is true regardless of what p is.* Such formulas that are never false are called *tautologies*.

When a proposition has the form of a tautology, it is, in a sense, meaningless.

Example. "I will wake up tomorrow, or I won't." This proposition is certainly *true*, but its truth has nothing to do with the subject matter. Rather, it is true *by its structure alone*: by the fact that it has the form of the tautology $p \vee \neg p$.

Still, it is sometimes useful to make a tautological statement.

Example. Suppose you want to prove some proposition q. To do so, you take an indirect approach: you take some other proposition p, and note that either p is true, or p is false. That is, $p \vee \neg p$. Then, you show that $p \implies q$ and $\neg p \implies q$, by assuming that p is true or false (respectively) and in each case deducing q. By doing so, you conclude that $(p \vee \neg p) \implies q$, by the equivalence $(p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$ which we discussed previously. Because $p \vee \neg p$ is tautologically true, you can then conclude q by modus ponens. This kind of proof is called *proof by cases*.

Notice that if we make a trivial* truth table for $\top$ itself, we obtain the same column:

p	$\top$
$\bot$	$\top$
$\top$	$\top$

Therefore, we can translate the statement that "$p \vee \neg p$ is a tautology" into propositional logic as "$p \vee \neg p \equiv \top$."† As a result, $\top$ stands not only for any true statement, but also for any tautological formula. Do Exercises 7.1–7.6.

Now that you know what it means to say that a formula is equivalent to $\top$ or $\bot$, I can provide you with more equivalences to mull over. But already I have grown tired of writing "Judge" before every exercise: therefore, I will omit this word from now on. That is, whenever an exercise simply contains a proposition without any explanation as to what to do with it, you are to *judge* it. Do Exercises 7.7–7.23.

The concept of tautology also allows us to further clarify the relationship between a logical equivalence and the corresponding biconditional. If F and G are two formulas, then:

$$(F \equiv G) \iff [(F \iff G) \equiv \top].^{\ddagger}$$

That is, to say that F and G are logically equivalent is to say that the biconditional between them is a tautology.§ Do Exercise 7.24.

Exercises

7.1 ○ Does this make sense?

7.2 ○ Are we *abusing notation*?[a]

7.3 ⓢ What would we probably mean by a "contradictory" formula?

7.4 ○ Find an example of such a contradiction.

7.5 ⓢ Using the symbol for logical equivalence, state that your contradiction is such.

7.6 ○ If *part* of a formula is contradictory, does that always render the entire formula contradictory?

7.7 ⓢ $p \wedge \bot \equiv \bot$.

7.8 ⓢ $p \wedge \top \equiv \top$.

7.9 ⓢ $p \wedge \top \equiv p$.

7.10 ⓢ $p \vee \top \equiv \top$.

7.11 ⓢ $p \vee \top \equiv p$.

7.12 ⓢ $p \vee \bot \equiv \bot$.

* In mathematics, we say things are "trivial" when they are really basic. The term is somewhat negative, but it is my opinion that trivialities can be very interesting.

† In fact, this is the formal statement of the Law of the Excluded Middle.

‡ We often use square brackets to avoid having too many parentheses right next to each other.

§ Notice how much more clearly we were able to understand this relationship after becoming familiar with the concept of tautology. This happens constantly in mathematics, and is one of its many pleasures.

7.13 ⓢ $p \vee \bot \equiv p$.

7.14 ⓢ $p \wedge (p \vee q) \equiv p$.

7.15 ⓢ $p \vee (p \wedge q) \equiv p$.

7.16 ⓢ $p \wedge \neg p \equiv \bot$.

7.17 ⓢ $p \implies p \equiv \top$.

7.18 ⓢ $p \implies \neg p \equiv \bot$.

7.19 ⓢ $p \implies \bot \equiv \neg p$.

7.20 ⓢ $p \implies \top \equiv p$.

7.21 ⓢ $\bot \implies p \equiv \top$.

7.22 ⓢ $\top \implies p \equiv \top$.

7.23 ⓢ $p \wedge \neg p \equiv p \implies \neg p$.

7.24 ○ Convince yourself that this is true, and verify it with a few examples.

[a] To "abuse" notation is to intentionally use it incorrectly for the sake of practicality.

*

Earlier, I said that the difference between a conditional and an English *if... then* statement is that the former merely claims that it is not actually *the case* that the premise is true and the conclusion false, while the latter claims that this state of affairs is *impossible*. In that sense, a logical implication is essentially a true *if... then* statement.

Specifically, logical implications correspond to those *if... then* statements which are *true by their structure alone*. There are also those true because of their *content*, and these cannot be expressed as logical implications. Later in this chapter, we will learn how such content-based *if... then* statements can be expressed in logic.

8 Logical implications

If a tautological biconditional is a logical equivalence, then what might a tautological conditional be?

Example. Consider the following truth table.

p	q	$p \wedge q$	$(p \wedge q) \implies p$
$\bot$	$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\bot$	$\top$
$\top$	$\bot$	$\bot$	$\top$
$\top$	$\top$	$\top$	$\top$

Observe that this is a tautological conditional whose premise is $p \wedge q$ and whose conclusion is p. Such a tautology is called a *logical implication*, and we say that $p \wedge q$ *logically implies* p, or more concisely that $p \wedge q \models p$.

We can see that a conditional is tautological when and only when it is impossible for its conclusion to be false while its premise is true.* There are three ways this can happen:

1. It is impossible for the premise to be true. E.g., $(p \wedge \neg p) \implies q$.
2. It is impossible for the conclusion to be false. E.g., $p \implies (q \vee \neg q)$.
3. The premise somehow claims that the conclusion is true. E.g., $(p \wedge q) \implies p$.

Of these, the first two ways are fairly straightforward: if the premise is contradictory, or the conclusion is tautological, the conditional between them will always be true. The third requires first that the premise and conclusion share certain variables, so that the premise can contain information about the conclusion, and second that this information actually implies the conclusion.

The first of these requirements (i.e., the sharing of variables) is easy to understand. The second is more complicated, and I'll just give you a couple of illustrations.

In the case of the truth table above, it is not possible for $p \wedge q$ to be true while p is false, because if p were false then $p \wedge q$ would be false. Therefore, the single circumstance in which the conditional might be falsified has been ruled out.

In contrast, within the formula for the conditional operator itself (i.e., $p \implies q$) there are no variables shared between the premise and conclusion, and therefore no way for the premise to *affect* the conclusion in any way.

So, a logical implication is a tautological conditional, and a logical equivalence is a tautological biconditional. To further clarify the relationship between logical implication and equivalence, suppose you find two formulas F and G such that $F \models G$ and $G \models F$. Then $F \implies G$ is always true, and $G \implies F$ is always true, and therefore $(F \implies G) \wedge (G \implies F)$ is always true. In other words, $F \iff G$ is tautological, and hence $F \equiv G$. Therefore, the exact same relationship holds between logical implication and logical equivance as holds between the conditional and biconditional: $F \equiv G$ means $(F \models G) \wedge (G \models F)$, just like $F \iff G$ means $(F \implies G) \wedge (G \implies F)$.

We have seen how useful logical equivalences can be for purposes of deduction. Logical implication can be used similarly, but in a simpler and more limited manner. Where logical equivalences could be used to derive other equivalences (by three methods) or to deduce a new theorem from an old theorem, logical implications are generally used only to deduce theorems.*

Here is how it works. If we know a theorem with the form of some formula A, and we know that $A \models B$, then we can deduce the corresponding proposition with the form of B.

Example. We know that $p \wedge q \models p$. If we also know that $(2 + 2 = 4) \wedge (4 + 4 = 8)$, we can deduce that $2 + 2 = 4$.

Do Exercises 8.1–8.6.

*

They can also be used to deduce other logical implications, in the following manner: if $F \models G$ and $G \models H$, we can conclude that $F \models H$.[a] However, this is not particularly useful, so I relegate its mention to this note only. Still, take a moment to think about why it works.

[a] This kind of relation, where, if A is related to B and B is related to C, we can deduce that A is related to C, is called *transitive* (because the relation seems to move (i.e., transition) from A through B to C).

Exercises

8.1 ⓢ $p \models q$.

8.2 ⓢ $p \models p$.

8.3 ⓢ $p \models p \vee q$.

8.4 ⓢ $p \vee q \models p$.

8.5 ⓢ $p \wedge (p \implies q) \models q$.

8.6 ⓢ $(p \implies q) \wedge (q \implies r) \models (p \implies r)$.

9 Predicates and predicate logic

So far, we have discussed *propositional* logic, in which we ignore the contents of propositions and care only about their truth values. But to do mathematics, we need to actually say specific things about mathematical objects. That is, we need to consider the details of atomic propositions.

To understand how to form an atomic proposition in logic, let's look at the structure of one in English: "The firetruck is red." The sentence's grammatical structure is

$$[\texttt{Subject}][\texttt{Predicate}].$$

The subject is "The firetruck" and "is red" is the predicate.

That is, we combine a thing with a property to claim that the thing has that property. (Here the "thing" is the subject, and the "property" is the predicate.) In logic, we do something very similar. We first construct a predicate, and we use it to make a claim as to the objects of which that predicate is true. However, logical predicates look different than English predicates.

Example. "x is a set."

Essentially the only difference is that we write a variable as a placeholder for the subject of the proposition. We would not write "is red," but rather "f is red." This is nicer, in that it doesn't look like a sentence fragment and completely clarifies where the subject goes.*

Now that variables are involved, it is best to think of predicates as formulas: where in propositional logic we plugged in propositions to turn a formula into a proposition, now we plug in *mathematical objects* to turn a predicate into a proposition.

Example. If we take the predicate "x is a set" and plug in something that isn't a set, the predicate becomes a false proposition, $\perp$. If we plug in a set, it becomes $\top$.†

We can also construct composite predicates, by applying logical operators to simpler predicates.

Example. We can conjoin "x is a set" and "x is a number" to form the composite predicate "x is a set $\wedge$ x is a number."

If you assign a mathematical object to the variable of a composite predicate, it becomes a composite proposition.

Example. Plugging the object 2 into the variable in "x is a set $\wedge$ x is a number," we obtain the composite proposition "2 is a set $\wedge$ 2 is a number."

As I said, we have left propositional logic: now we enter *predicate logic*, which is what I have previously referred to as simply *logic*. Like propositional logic, predicate logic is a language: albeit a slightly more complicated language.

All the symbols of propositional logic are also valid within predicate logic, except for propositional variables:‡ truth values, operators, and the logical equivalence sign '$\equiv$'. What we've learned about these symbols, and the ideas they represent, is still true.

The new symbols of predicate logic fall into three categories:

1. Variables,
2. Mathematical terms, and

*

This will be increasingly important as our predicates grow more complex.

†

The act of plugging in a particular subject for the variable in a predicate is called *application*. We say we are *applying* the predicate to that subject. If the result is $\top$, we say that the predicate *applies to* the subject and that the subject *satisfies* the predicate.

‡

Since now we write actual propositions.

3. Quantifiers.

The new variables are just like propositional variables, except that they can represent *anything*, not only propositions.

By "mathematical terms" I mean things like "is a set" or "divisible by 2" or even "2" itself. These are mathematical (as opposed to logical) in the sense that their meaning is not specified by logic but has to be supplied by axioms or definitions within a particular mathematical theory.

As you have seen above, we combine variables with mathematical terms to form predicates. One might expect that now we are done: to make an atomic proposition, just make a predicate and plug in a mathematical term as the subject.

But suppose we plugged "2" into the predicate "x is a set," and obtained the proposition "2 is a set." Well? *Is* 2 a set? Who knows?

The problem here is that *all* the terms in the proposition are mathematical, and unless we've previously defined those terms in logically meaningful terms, they will be meaningless. So, eventually we will be able to meaningfully write such propositions, but for now we must find a way to write a proposition whose terms are not entirely mathematical but contain some logical symbols.* We do this using *quantifiers*.

10 Quantifiers

Quantifiers are used to convert a predicate into a proposition by proposing something about that predicate: namely, that the predicate applies to *all things* or to *at least one thing*.† There are two basic quantifiers:‡

1. The *universal quantifier*, whose symbol is $\forall$, is used to claim that a predicate is true of all objects. That is, it claims that the predicate is universally true.

2. The *existential quantifier*, whose symbol is $\exists$, is used to claim that there exists something (or possibly more than one thing) that satisfies the predicate.

Do Exercise 10.1.

Examples. Here are some examples of propositions formed using quantifiers, and their English equivalents.

1. $(\forall x)(x \text{ is a set})$ is literally pronounced "for every x, x is a set," which means "all things are sets."§

2. $(\forall g)(g \text{ is divisible by } 2)$ is pronounced "for every g, g is divisible by two," which means "all things are divisible by two."¶

3. $(\exists y)(y + 3 = 0)$ is pronounced "there exists a y such that y plus three equals zero," which means "there's an object that you can add to three to get zero."‖

4. $(\exists \beta)(\beta \text{ is a set})$** is pronounced "there exists a beta such that beta is a set," which means "there exists a set."

*

This explanation is admittedly vague. You will understand what I mean much more clearly in the next chapter, when we state the axioms from which all other propositions will inherit their meaning.

†

Quantifiers are so named because they make a claim about a quantity: namely, how many objects the predicate applies to.

‡

There is one other that is commonly used, and I will introduce it later in this chapter.

§

Here it is the symbol '$\forall$' that is pronounced "for every."

¶

Notice that this is a stronger claim than "all *numbers* are divisible by 2." More on this soon.

‖

Here it is the symbol '$\exists$' that is pronounced "there exists," and the comma that is pronounced "such that."

**

I recommend you go online and familiarize yourself with the Greek alphabet. You will feel much more at home in mathematics.

*

I was hoping to illustrate this by saying "for example, our proposition above is clearly false because the King of England is not a set." However, this might presume too much. Historically, it has not paid to underestimate set theory.

†

Even so, we are not yet in a position to prove that there exists a set. To do so, we need to establish axioms concerning sets.

‡

In a sense, the predicate is the subject of the proposition, and the quantifier is the true predicate.

Let's inspect one of these examples more closely.

$$(\forall x)(x \text{ is a set}).$$

This can be separated into two parts:

$$\boxed{(\forall x)}\ \boxed{(x \text{ is a set})}.$$

The right-hand part is a predicate, which we are already familiar with. The left-hand part is a *quantification*, which uses the universal quantifier to *quantify* the variable x.

A quantification (e.g., $(\forall x)$) is four symbols long:

1. The first is a parenthesis.
2. The second is a quantifier. In our examples above, it was the quantifiers that were pronounced as "for every" or "there exists."
3. The third is a variable.
4. The fourth is a closing parenthesis. In universal quantifications, this is pronounced as a comma. In existential quantifications, it is pronounced "such that."

The above quantification, $(\forall x)$, uses the universal quantifier, and therefore says "I am claiming that the following predicate is true for every single object x." This is a very strong claim: it can be disproven by a single counterexample,* and to prove it you must make the argument that no counterexample exists.

Alternatively, we could write:

$$(\exists x)(x \text{ is a set}).$$

This proposition is identical to the previous except that we use the existential quantifier, and therefore only claim that there is at least one x satisfying the predicate. This is a much weaker claim, and generally much harder to disprove and much easier to prove:† to disprove it, we must show that it is never true, but to prove it we need only provide a single example.

Do Exercise 10.2.

As I said before, to quantify a predicate is to state a proposition about that predicate.‡ Like all propositions, such quantified statements have truth values. Here is how we determine them:

1. A universally quantified statement is true only if its predicate applies to all objects. It is false if there exists any object that does not satisfy the predicate.
2. An existentially quantified statement is true if there exists any object that satisfies the predicate. It is false only if every last subject fails to satisfy the predicate.

Do Exercise 10.3.

Exercises

10.1 ⓢ Why do we use these symbols: $\forall$ and $\exists$?

10.2 ⓢ Translate the following propositions into predicate logic.[a]

1. There exists a unicorn.
2. There exists something red.
3. All things are red.

10.3 ⓢ Determine the truth values of the propositions in Exercise 10.2, and make sure your own thought process is fairly clear to you.

[a] Remember the role of "mathematical terms" in constructing predicates.

11 Quantifiers in composite propositions

In this section we discuss the application of logical operators to quantified predicates, and the quantification of composite predicates.

First, let's consider the denial of quantified predicates. You may have noticed that there is something existential about a false universal, and something universal about a false existential. We will now examine this precisely. Suppose P is some predicate in which x is the variable. Then we have the following relationships between the existential and universal quantifications of that predicate:

1. $\neg(\forall x)P(x)^* \iff (\exists x)\neg P(x).^*$
2. $\neg(\exists x)P(x) \iff (\forall x)\neg P(x).^\dagger$

In other words, to say that a predicate is not always true is to say that it is sometimes false, and to say that it is never true is to say that it is always false.

We have now seen how quantifiers react under denial: next, through a series of exercises you will investigate how quantifiers interact under conjunction and disjunction. Do Exercise 11.2.

Having worked through Exercise 11.2, you are now ready for another insight regarding quantifiers. In Exercise 11.2, we found that universal quantification distributes over conjunction (but not disjunction), and existential quantification distributes over disjunction (but not conjunction). These facts reflect hidden relationships between universal quantification and conjunction, and between existential quantification and disjunction.

Consider the following universally quantified proposition:

$$(\forall x)(x \text{ is red}).$$

That is, every object is red. Now, instead of using a quantifier, we could express the same thing by just exhaustively listing every individual object and claiming that each is red. In other words, where $x_1, x_2, \ldots$ etc. are the (possibly infinite) objects, we could say:

$$(x_1 \text{ is red}) \wedge (x_2 \text{ is red}) \wedge \ldots \text{ etc.}$$

*

If a variable P denotes a predicate, then the application of P to x is written $P(x)$ and pronounced "P of x."

*

On the right-hand side of this biconditional, we have applied a logical operator (namely denial) to a predicate. Do Exercise 11.1.

†

It is commonly believed that it is difficult to "prove a negative." This is wrong, but reflects a truth that most people simply lack the terminology to articulate.

Example. "He wanted me to prove that unicorns *don't* exist! How am I supposed to prove a negative?"

This difficulty arises not from negativity, but from hidden universality. "Unicorns exist" is an existential statement, so its denial is universal: to prove unicorns don't exist is to prove that all things are non-unicorns.

Example. Not all that glitters is gold.

This is a negative statement, but, because it is the denial of a universal, it is an existential. Hence it can be proven by finding a single example of something that glitters but is not gold.

That is, a universal quantification is equivalent to an infinite sequence of conjunctions. Now, let us return to the question of distributivity: Consider this conjunction of two universally quantified statements:

$$(\forall x)(x \text{ is red}) \wedge (\forall x)(x \text{ is heavy}).$$

But by the connection with conjunction that we just discovered, this becomes:

$$[(x_1 \text{ is red}) \wedge (x_2 \text{ is red}) \wedge \ldots \text{ etc.}] \wedge [(x_1 \text{ is heavy}) \wedge (x_2 \text{ is heavy}) \wedge \ldots \text{ etc.}]$$

Because conjunction is associative, we can remove the square brackets. Because it is commutative, we can rearrange the order. By doing both we can obtain the following:

$$(x_1 \text{ is red} \wedge x_1 \text{ is heavy}) \wedge (x_2 \text{ is red} \wedge x_2 \text{ is heavy}) \wedge \ldots \text{ etc.}$$

And now by reversing our universal/conjunction connection, we obtain

$$(\forall x)(x \text{ is red} \wedge x \text{ is heavy}).$$

That is, the distributivity of the universal quantifier over conjunction can be explained by its hidden conjunctive nature. Do Exercise 11.3.

We have now considered three out of the five logical operators with respect to their interactions with quantifiers. Now we will discuss the remaining two: first the conditional, then the biconditional.

Throughout this chapter, we have been approaching the English *if... then* statement, and now we have found it. To say that "$(\forall x)(P(x) \implies Q(x))$" is to say that any object having property P will *necessarily* have property Q. In other words, that it is *impossible* that any object could have P without Q.

This can only happen if there is some genuine relevance between the properties P and Q, such that having P *necessitates* having Q. Such a relationship can only hold between premise and conclusion as a result of the fundamental axioms defining the properties of the basic mathematical objects of the theory in question, and the definitions of the concepts involved in the predicates P and Q. That is, where the logical implication is true by structure alone, the universal conditional is content-justified.

When I first explained that the conditional is distinct from the English *if... then* statement, I said that, nonetheless, you should essentially think of it as one. Furthermore, in my solutions to exercises on equivalences regarding conditionals, I appealed to our familiar intuitions about *if... then* statements to explain results. One might reasonably wonder why I seem to stress a distinction and then urge its dismissal. The answer is that, for the rest of this book, the conditionals we encounter will almost exclusively be universally quantified, and in this context, our intuitions about *if... then* statements are valuable.

Do Exercises 11.4–11.5.

Exercises

11.1 ⑤ What does this do to the predicate?

11.2 ⑤ For each of the following biconditionals,[a] decide if you think it *should* be

true or not, based on your current logical intuition.[b] Then, read the solutions, and update your intuition accordingly. That is, *judge* the following.

1. $(\forall x)P(x) \land (\forall x)Q(x) \iff (\forall x)(P(x) \land Q(x))$.
2. $(\forall x)P(x) \lor (\forall x)Q(x) \iff (\forall x)(P(x) \lor Q(x))$.
3. $(\exists x)P(x) \land (\exists x)Q(x) \iff (\exists x)(P(x) \land Q(x))$.
4. $(\exists x)P(x) \lor (\exists x)Q(x) \iff (\exists x)(P(x) \lor Q(x))$.
5. $(\forall x)P(x) \land (\exists x)Q(x) \iff (\exists x)(P(x) \land Q(x))$.

11.3 ○ Work out the relation between existential quantification and disjunction.

11.4 ⓢ Same as in Exercise 11.2: first think, then update your intuitions.

1. $(\forall x)P(x) \iff (\exists x)P(x)$.
2. $(\forall x)(P(x) \implies Q(x)) \iff [(\forall x)P(x) \implies (\forall x)Q(x)]$.
3. $(\forall x)(P(x) \iff Q(x)) \iff [(\forall x)P(x) \iff (\forall x)Q(x)]$.
4. $(\exists x)(P(x) \implies Q(x)) \iff [(\exists x)P(x) \implies (\exists x)Q(x)]$.

11.5 ⓢ In Exercise 11.2, how important is it that all the variables are x? That is, how would things change if one mixed in some y's or β's?

[a] Why not logical equivalences? That is, why "$\iff$" rather than "$\equiv$"?

[b] It may be that one direction of the conditional holds, but not the other: that is, `left` $\implies$ `right` but $\neg$(`right` $\implies$ `left`). Therefore, make your intuitive assessments separately for each direction and then bring them together to form a prediction for the biconditional as a whole.

*

This is called *currying*, named after the great logician Haskell Curry. However, the idea was actually first published (as far as we know) in the work of the logician Moses Schönfinkel, in 1924. This kind of thing happens often in the sciences.

12 Multivariate predicates

So far we have spoken of predicates with a single variable: that is, *univariate* predicates. Since variables stand for the subjects of the sentence, univariate predicates are those that speak about a single subject. Now we will consider predicates, such as relations, that speak of more than one subject.

Example. "x is older than y."

Such predicates are called *multivariate*. To convert a multivariate predicate into a proposition, that is, to apply it, you must supply as many subjects as there are variables. Thus, we might apply the predicate above to the two subjects "my dog" and "the Himalayas," producing either "The Himalayas are older than my dog" or "My dog is older than the Himalayas," depending on which subject we insert into which variable. On the other hand, you could *partially* apply it and only supply one subject, producing another predicate like "The Himalayas are older than y."*

But as we will see when we get to the set axioms, this is not usually how we convert predicates into propositions: instead, we use quantifiers. How does this work, when the predicate has multiple variables?

Example. $(\forall x)(\exists y)(y \text{ is the mother of } x)$.* This is pronounced "For every object x, there exists an object y such that y is the mother of x." That is, every object has a mother.†

Here we have used two quantifiers, one nested inside the other. Suppose we had only used one:

$$(\exists x)(y \text{ is the mother of } x).$$

This is not yet a proposition, but is still a predicate: namely, one that claims that its subject is a mother. To convert any predicate to a proposition, all of its variables must be quantified.‡ This is well worth pondering.

This is a good time to discuss *quantifier scope*. First: an *instance* of a variable is a distinct drawing of that variable's symbol on the page.

Example. In "x is x," there are two instances of the single variable x.

Then the *scope* of a quantifier is the collection of variable instances that it binds. Until now, you might have assumed that a quantifier would bind every instance of the variable named in the quantification, but this is not the case. Rather, a quantification binds every instance of its variable within the first parenthesis group to the *right* of that quantification (that is not itself a quantification) and *within its parenthesis-group*.

Example. The expression "$(\forall x)(x \text{ is red}) \wedge (x \text{ is heavy})$" is not a proposition, but a predicate, because the universal quantification does not bind the right-hand instance of x. If you were to supply the subject "the firetruck" to this predicate, it would become the following proposition: "Everything is red, and the firetruck is heavy."

At this point you are familiar enough with quantifiers to be shown some of their more advanced uses. Earlier, I said that quantifiers are so named because they assert a quantity: namely, how many objects satisfy a given predicate. The existential corresponds to "some" and the universal to "all." But quantifiers can also be used to express several other quantities. Here are the ones I know of.

1. Some: $(\exists x)P(x)$.
2. None: $\neg(\exists x)P(x)$.
3. All: $(\forall x)P(x)$.
4. Not all: $\neg(\forall x)P(x)$.
5. Some but not all: $(\exists x)P(x) \wedge \neg(\forall x)P(x)$.
6. Exactly one: $(\exists x)(\forall y)(P(y) \iff y = x)$. That is, there is some object x such that satisfying P is equivalent to identity with x.

   This kind of proposition is so useful in mathematics that we establish a third quantifier to abbreviate them: the "unique existential quantifier," denoted $(\exists!)$.§ Therefore, we would write the above proposition as follows:

   $$(\exists! x)P(x).$$

   This is then pronounced "there exists exactly one x such that $P(x)$."¶

*

The order of the two quantifications—that is, $(\forall x)$ and then $(\exists y)$—is very important, as we will discuss in the exercises below.

†

Notice, the translation is not "every *person* has a mother": the claim is not restricted to people, but generalized to all objects, including, for example, rocks. To restrict the claim to *people*, we would have to write: $(\forall x)(x \text{ is a person} \implies (\exists y)(y \text{ is the mother of } x))$. That is, for any object x, *if* x is a person *then* it has a mother.

In this way, a conditional can qualify a statement, restrict its scope, weaken its claim. Our original claim would be false, because rocks don't have mothers.[a] But this qualified statement is arguably true.[b]

[a] Except metaphorically, perhaps.

[b] So long as one doesn't take its present tense too literally.

‡

A quantified variable is also called *bound*, and a non-quantified variable is called *free*. Hence, a predicate becomes a proposition when all of its variables are bound. While it has even a single free variable, it remains a predicate.

§

That is, we add an exclamation mark.

¶

Here the exclamation mark is pronounced "exactly one," and the closing parenthesis is pronounced "such that," as with the normal existential quantifier.

Do Exercises 12.1–12.2.

This concludes our discussion of logic—the language of mathematics, in whose propositions we state our axioms and theorems. In analogy with natural languages, we might say that you have taken a first course in logic, enough to haltingly read and write and speak. You are not yet truly fluent, but still you are as prepared as you can be: fluency in any language can only be achieved through *use*.

Exercises

12.1 ⓢ Is it true that $(\forall x)(\exists y)P(x,y)$[a] $\iff$ $(\exists y)(\forall x)P(x,y)$?

12.2 ⓢ Is it true that $(\forall x)(\exists y)P(x,y) \iff (\forall y)(\exists x)P(y,x)$?

[a]When we apply a multivariate predicate, we list all the subjects between the parentheses. Note that order may matter here: If P denotes "x is the mother of y," then $P(\texttt{Ulysses},\texttt{Mary})$ is very different from $P(\texttt{Mary},\texttt{Ulysses})$: the former indicates that Ulysses is Mary's mother.

13 Proof and rules of inference

However, before we move on to our particular mathematical theory of sets, there remains one important topic of more general applicability: the matter of inference. That is, now that we have discussed the language in which mathematical propositions are written, we should explain how we determine their truth.

First, of course, one needs to come up with a proposition. As mentioned before, mathematicians use their well-trained intuition to decide upon a promising proposition. But in mathematics, intuition is never enough. Instead, mathematicians require an indisputable *proof* that the proposition is actually true.

To illustrate how much social importance the mathematical community places upon proof, I need only reference the Millennium Prize problems: six mathematical *conjectures* (i.e., propositions) whose proofs are to be rewarded by the Clay Mathematics Institute with $1,000,000 (USD) each. Some of these conjectures are not only intuitively likely, but extremely well-verified experimentally, such that within physics they would be considered established fact; but within mathematics, this verification is merely a failure to disprove. Mathematicians value true, indisputable proof so highly that after all this evidence we are not satisfied but willing to pay millions of dollars to whomever can give it to us.

So, what exactly constitutes a "proof"? Essentially, it is an argument strong enough to convince the mathematicians of one's time. That standard has changed over the history of mathematics, and continues to change. But from the beginning, the most fundamental goal of proof has been to render it impossible for our conclusions to be false.

Currently, we demand a clear, written demonstration that the proposition in question can be derived from the tautologies of logic and the axioms of set theory by strictly following certain agreed-upon rules of inference. Those rules are written in the form:

$$\frac{\text{Premises}}{\text{Conclusion}}\ (\text{Rule}),$$

which indicates that, having determined the truth of the Premises, one can use the Rule to deduce the truth of the Conclusion. Without further ado, let me list our rules of inference.*

*

There are many other types of inference we will use, but all of those can be performed as larger arguments using these basic rules; this relationship is analogous to that between atoms and molecules, or between atomic propositions and composite propositions, or between primitive terminology and defined terminology.

1. $\dfrac{\text{Assume } \mathbf{P} \ldots \text{ deduce } \mathbf{Q}}{\mathbf{P} \implies \mathbf{Q}}$ (Conditional introduction).

   Here is what this means, in terms of its use in a proof:

   1. Say something like "Suppose $\mathbf{P}$" to enter into a hypothetical.
   2. Within that hypothetical, deduce that $\mathbf{Q}$ must be true.
   3. Observing that $\mathbf{Q}$ followed from the assumption of $\mathbf{P}$, we conclude that $\mathbf{P} \implies \mathbf{Q}$.

   Example. Suppose there is a law stating that anyone owning a dog must pet it no fewer than three times per day, and we know for certain that anyone who breaks this law will be successfully fined for \$20. In addition, suppose Guillaume owns a dog, and pets him exactly twice per day. Therefore, in this hypothetical, Gillaume is going to be fined \$20. From this reasoning we can conclude that, if there is such a law, and if Guillaume pets his dog exactly twice per day, then he will be fined \$20.

2. $\dfrac{\mathbf{P} \wedge (\mathbf{P} \implies \mathbf{Q})}{\mathbf{Q}}$ (Conditional elimination).

   This is simply modus ponens, and it allows us to make use of many logical implications for the purpose of deduction.

3. $\dfrac{\mathbf{P}(\mathbf{c}) \text{ where } \mathbf{c} \text{ is an arbitrary constant}}{(\forall x)\mathbf{P}(x)}$ (Universal introduction).

   This works as follows:

   1. Let $\mathbf{c}$ be an arbitrary object.
   2. Deduce some fact about $\mathbf{c}$, denoted $\mathbf{P}(\mathbf{c})$.
   3. Therefore, because we didn't assume *anything* about $\mathbf{c}$ and yet managed to deduce $\mathbf{P}(\mathbf{c})$, we can conclude that $\mathbf{P}(x)$ must be true *for all* x.

   Example. Let w be an arbitrary object. Tautologically, either w is a set or it isn't. Therefore, every object either is or is not a set.

4. $\dfrac{(\forall x)\mathbf{P}(x)}{\mathbf{P}(\mathbf{y})}$ (Universal elimination).

   That is, if $\mathbf{P}(x)$ is true for every x, it must therefore be true for $\mathbf{y}$ in particular (whatever $\mathbf{y}$ may be).

   Example. As we found in the previous example, every object is either a set or a non-set. Therefore, either I am a set, or I am not a set.

5. $\dfrac{\mathbf{P}(\mathbf{y})}{(\exists x)\mathbf{P}(x)}$ (Existential introduction).

   That is, if a predicate $\mathbf{P}$ applies to some specific object $\mathbf{y}$, then this is an example proving that there is *some* object to which $\mathbf{P}$ applies: namely, $\mathbf{y}$.

   Example. I have red hair. Therefore, there exists an object with red hair.

6. $$\frac{(\exists x)\mathbf{P}(x) \qquad \text{Assume } \mathbf{P}(\mathbf{c}) \text{ (for a fresh } \mathbf{c}) \ldots \text{ deduce } \mathbf{Q}}{\mathbf{Q} \quad \text{(where } \mathbf{c} \text{ does not appear in } \mathbf{Q})} \text{ (Existential elimination).}$$

An existential statement claims that there is some object with the given property. The purpose of the existential elimination rule is to be able to give a temporary name to one of those objects for the purpose of reasoning. At the end of this explanation, I will explain why it is necessary to stipulate that $\mathbf{c}$ does not appear in $\mathbf{Q}$.

1. We start with the knowledge that $(\exists x)\mathbf{P}(x)$. That is, there is at least one object satisfying the predicate $\mathbf{P}$.
2. Therefore, we say "Let $\mathbf{c}$ be one such object." In other words, we define a temporary name for one of the objects satisfying $\mathbf{P}$, for use within the proof.
3. Now deduce some useful conclusion, denoted $\mathbf{Q}$. As long as $\mathbf{Q}$ doesn't mention the name $\mathbf{c}$, you may conclude it.

Example. We know there exists an object with red hair. For the purposes of this discussion, let "Obama" be one such object. Now, as we have found before, all objects are either sets or non-sets: therefore, Obama is a non-set. Conclusion: by existential introduction, there exists a non-set with red hair. Because our conclusion did not mention Obama, it remains valid. But if our conclusion was instead: "*Obama* is a non-set with red hair," then we might think ex-President Barack Obama has red hair, which would be incorrect.

So, a mathematician first uses their intuition to select a likely proposition. Then they select a collection of premises—axioms and previously proven theorems—which they think are sufficient to prove this proposition, and use the rules of inference to deduce the proposition, step by step, from these premises. How exactly do they know which premises are relevant, and which rules they should apply, and how to apply them in each specific case? That is, how do you design a proof?

The answer is that proof is an art, and it is learned primarily through experience. and until you have obtained this experience I can only provide the following vague gesture: your initial intuition as to why the proposition is probably true will guide you to an extent as to how to prove it. With experience, your intuition will be trained to adhere more closely with the rigors of proof, so that by the time you have decided on a proposition, you will have some sense of how to prove it.

In addition, there are certain patterns of deduction, certain so-called *proof techniques*, which can be employed as tools in your inferential toolbox. As we go, I will introduce these patterns and explain the types of situation in which each one is useful. Throughout your study of this book you will find plenty of opportunity to observe high-quality examples of proofs and come up with your own. In many of my examples of proofs, I will explain why I choose each technique in each particular circumstance.

Of course, it is also possible that your initial intuition is wrong, and, in the midst of writing your proof you find you have to switch to disproof. Or, perhaps you finish writing your proof, and find later that the "proven" theorem starts contradicting your other results, and in retrospect you find that you made a mistake in your proof.

So, in short, you'll figure it out! All of this will become somewhat more clear in Chapter 3, where I provide the first real example of proof, and much more clear again in Chapter 4, where we will work through a number of very simple proofs together.

14 Conclusion

Let me summarize what we've discussed so far:

- In mathematics, we choose an idea to investigate. A particularly useful choice is the idea of a *set*.
- Having chosen an idea, we must capture its essence in a set of assumptions, called *axioms*. From these axioms we deduce further facts, called *theorems*.
- Logic allows us to precisely express axioms and theorems as *propositions*: claims of objective fact.
- Propositions are formed by combining variables, mathematical terms, and logical operators into a predicate, and then quantifying all the variables.
- Variables are letters serving as placeholders for the mathematical objects being spoken about.
- Variables are then combined with made-up words to form basic predicates: templates for claims about mathematical objects.
- Basic predicates can be combined into composite predicates using the logical operators.
- To form a proposition from a predicate, we quantify all of its variables. The quantifiers are the universal (denoted $\forall$), the existential (denoted $\exists$), and the unique existential (denoted $\exists!$).
- Once we have a proposition, we can construct *composite propositions* using the logical operators: denial ($\neg$), conjunction ($\wedge$), disjunction ($\vee$), the conditional ($\implies$), and the biconditional ($\iff$).
- The axioms and theorems of set theory will be formed in exactly this manner. Once it is formed, every proposition has a *truth value*—truth ($\top$) or falsehood ($\bot$)—determined by the tautologies of logic, the axioms of set theory, and the six rules of inference.
- Through exercises we have learned many properties of the logical operators and quantifiers, and used these to improve our logical intuitions. This gives us the basic familiarity required to understand the axioms and what follows from them.

Now, it is time to move on from logic and enter our study of mathematics proper, beginning with set theory. Do Exercise 14.1.

Exercises

14.1 ○ Remember to check the list of incorrect solutions at the back of the book.

Part I

Sets

*

This collection is empty.

Chapter 2

Introduction to set theory

1 Sets represent collections

Sets are mathematical objects designed to represent collections of things. To understand why we choose the set-theoretic axioms we do, let us first discuss the intuitive idea they are intended to capture.

Examples. Here are some examples of collections.

1. The objects in my safe.
2. Those World Chess Champions who have been President of the United States.*
3. Numbers.
4. Cats who live in New York City.

Such collections are very familiar from everyday life, and it is our good fortune that mathematics can be founded on their representatives.

2 Containment and defining predicates

Given any thing x and a collection C, either x is in C (denoted $x \in C$) or it isn't (denoted $\neg(x \in C)$). In the first case, we say that x is an *element*, or *member*, of C, or that x *belongs to* C, or that C *contains* x, etc.

To determine which things are in C, it is defined by a certain predicate. Then C contains x if and only if its defining predicate applies to x.

Example. For the collection of cats who live in New York City, the defining predicate is as follows: "x is a cat $\wedge$ x lives in New York City." My cats, who live in Prince Edward Island, Canada, fail this test and therefore are not members of that collection.

3 Intension and extension

The relationship between a collection and its defining predicate is best explained with language borrowed from philosophy. In philosophy, every concept is described as having an *intension* and an *extension*.

The extension of a concept is the group of all objects falling under it, while its intension is that which all its objects have in common: its defining characteristic.* In such terms, a collection is often thought of as the extension of some concept whose intension is clearly described by that collection's defining predicate.†

*: For our upcoming purposes, notice that an intension is easier to store in one's mind than an extension: one can easily grasp the essential idea (i.e., the intension) of a cat, but cannot hold simultaneous knowledge of every individual cat.

†: It is because of this close connection between collections and concepts that set theory serves as a foundation for mathematics. As mentioned at the start of Chapter 1, mathematics is the investigation of certain very clear concepts, called *mathematical* concepts. These mathematical concepts can each be defined as a certain set representing the extension of that mathematical concept. Therefore, by investigating certain sets we can study any mathematical concept we may be interested in, and so set theory serves as a proxy for mathematics in general.

4 The axiom of extensionality

We will now turn from the intuitive idea of a collection to the choice of axioms by which we hope to capture their essence within the mathematical concept of a set.

The only difference between intuitive everyday collections and mathematical sets (aside from the basic difference between any representation and what it represents‡) is that the identity of a set is determined only by what it contains. In contrast, two collections with the same contents might be considered to differ with regard to what is doing the containing (such as a box or a nation), or the arrangement of their contents, etc. Two such collections, therefore, would be mathematically represented by the same set in spite of their differences.§

‡: Such as, for example, the difference between a portrait and the actual person portrayed.

§: As a result, sets are more abstract than the collections they represent.

This motivates our first axiom of set theory, tacitly assumed by Cantor and later made explicit: the *axiom of extensionality*.

Axiom (Extensionality)**.** $(\forall A)(\forall B)(\forall x)[(x \in A \iff x \in B) \implies (A = B)]$.¶

That is, "if two sets have the same elements, they're identical." Do Exercise 4.1. Or, in other words: sets are nothing more than extensions.

Exercises

4.1 ○ Convince yourself that this is the correct translation from logic to English.

5 Abstraction and (the axiom of) comprehension

To move from extension to intension is called *abstraction*. It is in this sense that mathematics is often called *abstract*: from the objects of everyday experience we have abstracted certain concepts, and then abstracted further concepts from these first concepts, and so on. Another word for abstraction is *generalization*. To generalize an idea is to realize that there is a more general (or abstract) idea of which it is a specific instance.

Example. By observing in nature various approximately circular things, such as the sun and moon, we generalize to the more abstract idea of a circle that is not approximate but *ideal*, or *platonic*. After obtaining several similar ideas, such as those of squares and triangles, we abstract further to the idea of a *shape* in general.

Through abstraction, we obtain widely applicable concepts.

Example. The idea of a circle applies to all approximately circular objects in reality, as well as infinitely many imaginary circles. It even applies through metaphor to "circular reasoning" etc.

Conversely, to move from intension to extension is called *comprehension*. This motivates the *axiom schema of comprehension*,‖ which, together with the axiom of extensionality,

served as the (sometimes tacit) foundation for the early set theories of Cantor, Frege, and others. It states that for every predicate there is a set of exactly those objects to which that predicate applies.

Axiom schema (Comprehension)**.** *For every predicate* $\mathbf{P}$,* *it is an axiom that:*

$$(\exists S)(\forall x)(x \in S \iff \mathbf{P}(x)).$$

This assumption is not called simply an "axiom" but an "axiom schema" because every choice of the predicate $\mathbf{P}$ results in a distinct axiom. Rather it is a template, or *scheme*, from which infinitely many axioms can be produced. These axioms, together, are so powerful that every single mathematical truth follows from them. Unfortunately, they also entail some other rather dubious propositions.

¶

Here, the identity symbol = means "is identical with".

Generally, this symbol and the idea of identity (or *equality*) which it represents are taken to be part of logic itself, not mathematics proper. In that context, to say that $A = B$ is to say that mentions of A and B are interchangeable in any proposition.

In this book, I have instead chosen to *define* identity within mathematics and to *prove* that when two sets are identical (as so defined) they are therefore interchangeable, and that if they are interchangeable then they are identical (as so defined). This will be performed in the next chapter.

‖

The axiom schema of comprehension is sometimes (more often historically) misleadingly called that of *abstraction*.

6 Russell's paradox

A *paradox* is a situation in which something seems true but simultaneously false.

Example. In Chapter 1, we introduced the Liar's Paradox produced by the sentence "this statement is false." Seemingly, the Liar's Sentence must be both true and false.

When we find a paradox, we know we are somehow mistaken: it *seems* to us that some proposition is both true and false, but that cannot be. Therefore every paradox is a gift, for it invites us to correct an error in our thinking and thereby grow stronger.

Several paradoxes result from the axiom schema of comprehension; for now, we will discuss only one, known as *Russell's paradox*.† Consider the following predicate: $\neg(x \in x)$. That is: x doesn't contain itself. From the axiom schema of comprehension, we obtain the following axiom:

Axiom (Russell)**.** $(\exists R)(\forall x)(x \in R \iff \neg(x \in x))$.

That is, there exists a set, which we call R, containing all things that do not contain themselves. In other words: for every object x, R contains x if and only if x does not contain x. But since this is true for every object x, it must be true for R itself:

$$(\exists R)(R \in R \iff \neg(R \in R)).$$

That is, R contains R if and only if R does *not* contain R. Therefore, there exists a set whose self-containment is self-denying: the claim "$R \in R$" is the set-theoretic Liar's Sentence. Because negation flips the truth value of its argument, it is impossible for a proposition to have the same truth value as its negation: thus, on the grounds of merely the axiom schema of comprehension, we have proven a falsehood. Do Exercise 6.1.

Therefore, anyone assuming the axiom schema of comprehension places themself in a position of paradox. To resolve the paradox, they must either doubt their derivation of Russell's axiom given above, or they must doubt the axiom schema from which it was derived.

In general, it is entirely reasonable to doubt one's supposed "proofs", because humans regularly make logical missteps. This one, however, is so simple, and has been proven by so many different humans and indeed, later, by computers, that it is very likely correct.

In assuming the axiom schema of comprehension, early set theorists clearly assumed too much: there are some concepts with no coherent extension. Therefore, these early theories are dubbed "*naïve set theories*," because they are intuitively appealing but wrong. To establish a firm foundation for mathematics, we need more restricted axioms that are still intuitive and still powerful enough to prove the propositions we want to prove.

Exercises

6.1 ⓢ Why is the presence of this contradiction a serious problem, or why is it not?

7 ZFC

Over roughly twenty years, this was accomplished by the research of several mathematicians and logicians, culminating in what is now called *Zermelo-Fraenkel set theory with the axiom of choice*, or, more briefly, *ZFC*. This theory consists of seven axioms and two axiom schemata:*

1. Extensionality
2. Separation (schema)
3. Union
4. Pairing
5. Regularity
6. Power set
7. Infinity
8. Replacement (schema)
9. Choice

8 Grammar of set theory

Recall in the last chapter I mentioned that predicates are formed using "mathematical terms" and that these are provided by the particular mathematical theory of interest, rather than logic itself. We have now encountered the only *primitive* term† necessary for our study of mathematics—the set membership symbol $\in$—and are finally in position to formally specify the primitive grammar of set theory. To do so, I will state seven rules which, taken together, define the *primitive propositions* of our theory:

1. A *variable* is any italicized character of the Latin or Greek alphabets, with or without subscripts or superscripts: a, b, c, $\ldots$, X, Y, Z, α, β, γ, $\ldots$, Ψ, Ω, m_i, h_4, g^3, etc.
2. If $\mathbf{v}$ and $\mathbf{w}$ are variables,‡ then $\mathbf{v} \in \mathbf{w}$ is a primitive expression.

*

Let me explain why **P** is written in boldface.

Propositions within our theory—that is, those regarding sets—are called *theoretical* propositions. Propositions *about* our theory are called *metatheoretical*.[a]

Within a theoretical proposition, each variable stands for a set. Such variables, standing for the mathematical objects under investigation, are called theoretical variables.[b] With metatheoretical propositions, we speak of things other than sets: for example, predicates. Variables standing for such things are called metatheoretical variables,[c] and within this book they are written in boldface.

So this is why **P** is written in boldface: it is a metatheoretical variable standing not for a set but a predicate. Therefore, the axiom schema of comprehension is not a theoretical proposition about sets, but a metatheoretical proposition *about set theory*: specifically, it makes a claim about which theoretical propositions are axioms.

[a]Theoretical propositions are governed by a strict grammar particular to the theory in question, while metatheoretical propositions may be written freely. The grammar for our set theory is given below, in Section 8.

[b]Often abbreviated to "variable."

[c]Often abbreviated to "metavariable."

†

We will discuss some of the others later, when we have built up the prerequisites.

3. If $\mathbf{P}$ is a primitive expression, then $\neg\mathbf{P}$ is a primitive expression.
4. If $\mathbf{P}$ and $\mathbf{Q}$ are primitive expressions, then $(\mathbf{P} \wedge \mathbf{Q})$ is a primitive expression.
5. If $\mathbf{v}$ is a variable symbol, and $\mathbf{P}$ is a primitive expression, then $(\forall\mathbf{v})\mathbf{P}$ is a primitive expression.*
6. Nothing is a primitive expression unless stated to be so by rules 2–5.
7. A *primitive proposition* is any primitive expression whose variables are all bound.†

Do Exercise 8.1.

Exercises

8.1 ◯ Use these rules to construct an interesting primitive proposition that you think might be true.

9 Our study of mathematics

Before we begin our mathematical investigations, I would like to explain a little bit more about how they will proceed. Generally, we will repeat the following routine:

1. Define a new mathematical concept. Optionally, this may be accompanied by a philosophical explanation of the intuitive idea that concept represents.
2. Mathematically investigate that concept, and its relations with previous concepts, by generating and then *judging* various propositions about it.

Until we move on from mathematics to physics, each Part of this book focuses on a particular subfield of mathematics. This one is dedicated to set theory, while another, for example, is dedicated to geometry: the mathematical investigation of space.

For most of this book, each Part is divided into chapters, each of which focuses in turn upon a certain broad topic within its subfield. For example, in the Part on geometry, one chapter might focus on trigonometry: the mathematical study of the relations between the sides and angles of triangles. Within each chapter there are *sections*, most of which begin with the definition of a new concept. The remainder of each section will be dedicated to the investigation of its concept.

This very first Part, on set theory, is divided slightly differently, because we need to state *axioms* as well as definitions. In each chapter, we investigate sets as well as we can with the axioms we have, and when we can go no farther we begin the next chapter and introduce a new axiom. Therefore, each chapter of this Part is distinguished not by its topic, but by the additional axiom it depends upon. In this way, later chapters deal with subjects that depend upon a greater number of axioms. There are ten chapters on set theory, of which this is the first. Each of the other nine focuses on one of the axioms of ZFC, beginning, in the next chapter, with the axiom of extensionality.

*

I will not give their full statements here, because they involve terminology you are not yet familiar with. Nor will I try to express them in English, because any understanding you might thereby gain would be faulty. Suffice it to make these four observations, which you may verify going forward:

1. The axiom of extensionality remains unaltered.
2. The axiom schema of separation is a restriction of that of comprehension.
3. As far as we know, this restriction successfully removes all paradoxes from set theory. However, it is *too* restricted, on its own, to derive all of mathematics as its consequences.
4. The other seven axioms serve to carefully add back what was lost without introducing or reintroducing any paradoxes.

†

"Primitive" is in contradistinction to "defined": all defined terminology will be defined ultimately in terms of the primitive terminology we establish here.

‡

Here $\mathbf{v}$ and $\mathbf{w}$ are metavariables standing for theoretical variables: thus $\mathbf{v}$ might stand for the symbol x or y or any other symbol as specified in rule 1.

*

Notice that we have not included disjunction in these rules, nor the conditional or biconditional, nor the existential or unique existential quantifiers. This is because, as we found in the last chapter, these can all be defined in terms of negation, conjunction, and the universal quantifier.[a]

[a] In fact, all of the logical operators can be defined in terms of a single operator called `NAND` (short for "not and"), which I prefer to call *exclusion*. The exclusion of two propositions **P** and **Q** (note the boldface) is denoted $(\mathbf{P} \mid Q)$ and defined as follows:

P	**Q**	$(\mathbf{P} \mid \mathbf{Q})$
$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\top$
$\top$	$\bot$	$\top$
$\top$	$\top$	$\bot$

That is, $(\mathbf{P} \mid \mathbf{Q})$ claims that **P** and **Q** aren't both true: at least one of them is false. In other words, it claims they are mutually exclusive. Then negation can be defined like so:

$$\neg\mathbf{P} \text{ means } (\mathbf{P} \mid \mathbf{P}).$$

And with negation in hand, we define conjunction:

$$(\mathbf{P} \wedge \mathbf{Q}) \text{ means } \neg(\mathbf{P} \mid \mathbf{Q}).$$

Therefore, if it appeals to you (as it does to me), you may consider exclusion to be the only primitive logical operator.

†

Recall that a variable is *bound* within an expression if each of its instances are within the scope of some quantifier that binds it, and that a quantifier's scope reaches to the right as far as the nearest parenthesis-group.

*

An *identity* is a claim that "two things" are identical.

Example. "The moon = the moon" is an identity.

Chapter 3

Extensionality

1 Overview

In the previous chapter, we encountered an early formulation of the axiom of extensionality, which stated that if two sets have the same elements, they are identical:

Axiom (Extensionality)**.** $(\forall A)(\forall B)(\forall x)[(x \in A \iff x \in B) \implies (A = B)]$.

Unfortunately, the above formulation assumes familiarity with a primitive *logical* notion of identity indicated by the symbol $=$, and in my experience this assumption is unwarranted. Instead, we will:

1. Discuss the intuitive idea of identity.
2. Mathematically define the identity symbol $=$.
3. State a simplified version of the axiom of extensionality.
4. Use the definition and axiom to prove that our mathematical concept of identity has all the properties we intuitively expect.

2 Idea of identity

Like many English words, "identical" means slightly different things in different contexts. In this context, we mean "exactly the same thing." Therefore we say that two things are *identical* if and only if they are not in fact two things, but are rather the same thing. There must be absolutely no difference of any kind between them. Hence 2 is *only* identical to 2, or, more briefly, $2 = 2$. Of course, identity need not apply only to numbers: the moon = the moon.

Now, if all of our identities* were of the form $\mathbf{v} = \mathbf{v}$, they would be of little use. Much more commonly, we want identities of the form $\mathbf{v} = \mathbf{w}$, where $\mathbf{v}$ and $\mathbf{w}$ are two different expressions representing the same thing. Hence we might say "$2 = 1 + 1$," since '2' and '$1+1$' are expressions that denote the same mathematical object. We could not say "$2 = 3$," because the symbols '2' and '3' represent different mathematical objects.

*

Recall from chapter 1 that a free occurrence of a variable is one that is not bound by any quantifier.

As another example, we might say that "the star that rises at dawn = the star that sets at dusk." And this is a genuinely useful observation about the sun. However, it is not quite a *true* observation. By the time the morning star has become the evening star, it has done a lot of burning, and this changes it so that there are *differences* between the star that rises in the morning and the star that sets in the evening. Therefore, we cannot really consider them identical, even though colloquially we would consider them the same star. In the same way, you are not identical to your baby-self, even though most people would say your "identity" persists as you age. On the other hand, it would be entirely correct to say that "Superman = Clark Kent," so long as you aren't referring to him at one time on the left and a different time on the right.

As you can see, the notion of mathematical identity can be applied to everyday objects, but it is more strict than our everyday idea of identity. To summarize: each object is only ever identical to itself, and we write $\mathbf{v} = \mathbf{w}$ when $\mathbf{v}$ and $\mathbf{w}$ are two names for the same object.

Thus, when we try to determine if an identity is true, we are engaging in *semantics*: we are asking, "Do these two expressions really refer to exactly the same object? That is, do they *mean* exactly the same thing? Or are there any differences?"

3 Substitution and synonymy

For the purposes of mathematical investigation, it is useful to establish identities because, if we know that $\mathbf{v} = \mathbf{w}$, that means we can replace $\mathbf{v}$ with $\mathbf{w}$ or vice versa wherever we want within any proposition without changing the truth value of that proposition.

To be more precise: if n is a positive whole number and $\mathbf{P}$ contains n or more free occurrences of the variable $\mathbf{v}$, let "$\mathbf{P}(\mathbf{v}_n \mapsto \mathbf{w})$" denote the result of taking $\mathbf{P}$ and replacing the nth free occurrence of $\mathbf{v}$ (from the left) with the variable $\mathbf{w}$, provided that the resulting occurrence of $\mathbf{w}$ would be free.* If the resulting occurrence of $\mathbf{w}$ would be bound, or if $\mathbf{P}$ has fewer than n free occurrences of $\mathbf{v}$, then $\mathbf{P}(\mathbf{v}_n \mapsto \mathbf{w})$ denotes $\mathbf{P}$ unchanged.

Examples. If $\mathbf{v}$ denotes 'x', $\mathbf{w}$ denotes 'y', and $\mathbf{P}$ denotes "$x + x = z$," then:

1. $\mathbf{P}(\mathbf{v}_1 \mapsto \mathbf{w})$ denotes "$y + x = z$."
2. $\mathbf{P}(\mathbf{v}_2 \mapsto \mathbf{w})$ denotes "$x + y = z$."
3. $\mathbf{P}(\mathbf{v}_0 \mapsto \mathbf{w})$ is simply undefined.
4. $\mathbf{P}(\mathbf{v}_3 \mapsto \mathbf{w})$ denotes "$x + x = z$."

Examples. If $\mathbf{P}$ denotes "$(\forall x)(x = y)$", $\mathbf{v}$ denotes 'x' and $\mathbf{w}$ denotes 'y', then:

1. $\mathbf{P}(\mathbf{v}_1 \mapsto \mathbf{w})$ simply denotes $\mathbf{P}$.
2. $\mathbf{P}(\mathbf{w}_1 \mapsto \mathbf{v})$ also denotes merely $\mathbf{P}$.

Then we metamathematically define *synonymy* as follows:

Metadefinition 1. *"$\mathbf{v}$ is synonymous with $\mathbf{w}$" stands for "For every proposition $\mathbf{P}(\mathbf{v})$ mentioning $\mathbf{v}$, $\mathbf{P} \iff \mathbf{P}(\mathbf{v}_n \mapsto \mathbf{w})$ for each n."*

In other words, two expressions are synonymous if and only if you can replace one with the other wherever you want within any proposition without changing the truth value.

Returning to identity now, we can see that if $\mathbf{v} = \mathbf{w}$, then $\mathbf{v}$ is synonymous with $\mathbf{w}$. Furthermore, this property is actually *characteristic* of identity. That is, the converse holds: if two expressions are names for the same object, then the object(s) they represent must be identical. Therefore, synonymy is logically equivalent to identity: $(\mathbf{v} = \mathbf{w}) \iff (\mathbf{v}$ is synonymous with $\mathbf{w})$. This provides us with an easy way to test any proposed mathematical definition of identity: can we prove that, if two objects are "identical" (according to the proposed definition), does that necessarily render them synonymous? And conversely, if they are synonymous, does that guarantee they are "identical"?

*

Note that our official axiom is *numbered*, while our previous, unofficial axioms were not. It has been, and will continue to be, useful to discuss axioms which will not actually make it into our official theory. Thus, to avoid confusion, only our official list of nine axioms will be numbered. That way, you can look at whether an axiom is numbered to tell if it's official.

4 Definition of identity

Now that we have a clear intuition, here is our proposal for a set-theoretic definition of identity.

Definition 1 (=)**.** $\mathbf{v} = \mathbf{w}$ *stands for* $(\forall x)(x \in \mathbf{v} \iff x \in \mathbf{w})$.

That is, "to say that two objects are identical is to say that they have the same elements." The intuitive justification for this definition is that our theory is a theory of sets, and sets are so abstract that we care only about their contents: two sets are identical so long as they contain exactly the same objects.

The wisdom of this definition remains to be seen. We will have to investigate our new mathematical concept, and see if its properties align with our intuitions of identity, or if perhaps our intuitions need to be improved.

5 Axiom of extensionality

Now we state our first official axiom, which you will see is necessary for our defined concept of identity to behave as expected.

Axiom 1 (Extensionality)**.** $(\forall x)(\forall y)(x = y \implies (\forall z)(x \in z \iff y \in z))$.*

That is, if two sets have the same elements, then they are contained within the same sets. Note that, within the above axiom, we have used the defined symbol =. Therefore, we could replace this axiom with an equivalent one stated in more primitive language, as follows:

Axiom (Primitive extensionality)**.** $(\forall x)(\forall y)((\forall z)(z \in x \iff z \in y) \implies (\forall z)(x \in z \iff y \in z))$.

Here we have simply replaced "$x = y$" with its definition. As you can see, the primitive version extends off the page, but we are able to be much more concise by defining short expressions to stand for longer ones. As we progress in our investigations from simpler to more complex subjects, we will need to constantly define new mathematical concepts, and corresponding terminology, to combat the increasing complexity and length of our theorems. This is why, throughout our investigation, we will tend to go back and forth between defining terminology and proving theorems: before we discuss the theorem we need to build up the

appropriate vocabulary. Once we define the necessary terminology, we can prove some theorems using that terminology, and then we have to go back to definitions before moving on again.

6 Proof of synonymy

Now we want to prove that if $\mathbf{v} = \mathbf{w}$ then $\mathbf{v}$ and $\mathbf{w}$ are synonymous, and that if $\mathbf{v}$ and $\mathbf{w}$ are synonymous then $\mathbf{v} = \mathbf{w}$. In other words:

Metatheorem 1 (Synonymy)**.**
For any two variables $\mathbf{v}$ *and* $\mathbf{w}$, $(\mathbf{v} = \mathbf{w}) \iff (\mathbf{v}$ *is synonymous with* $\mathbf{w})$.*

Now that we have stated this (meta)theorem, we need to prove it. When planning for a proof, you should look at the theorem you're proving.

First, we see "For any two variables $\mathbf{v}$ and $\mathbf{w}$..." Therefore, we will need to start our proof by introducing those metamathematical variables $\mathbf{v}$ and $\mathbf{w}$, and indicating that the only thing we are assuming about the objects they represent is that they are variables (note: not *meta*variables).†

Next we see the $\iff$ symbol: we are trying to prove a biconditional statement. Therefore, we will have to prove two different things: that the premise implies the conclusion, and that the conclusion implies the premise. You will see that the right-to-left direction will be easy, but the left-to-right direction requires some preparation.

To prove the left-to-right conditional, we will need to assume that $\mathbf{v} = \mathbf{w}$ and thereby deduce that $\mathbf{v}$ is synonymous with $\mathbf{w}$ according to the definition of synonymy previously discussed. To prove that $\mathbf{v}$ and $\mathbf{w}$ are synonymous, we need to show that, within *any* set-theoretic proposition, we can swap one for the other without changing the truth value. Therefore, we should look for useful information about our set-theoretic propositions.

Referring to our grammar from Chapter 2, we see that propositions are a kind of primitive expression, and that primitive expressions are defined *inductively*: that is, there are basic expressions (of the form $\mathbf{x} \in \mathbf{y}$) given by rule 2, and rules (3–5) allow us to construct more complex expressions from simpler, existing ones (via logical operators and quantifiers).

In general, an inductive definition starts with *basic clauses*, which establish the simplest examples of the objects being defined (called *basic objects*).

Example. In our definition of primitive expressions found in the grammar in Chapter 2, rule 2 is our basic clause.‡ The basic objects are therefore expressions of the form $\mathbf{x} \in \mathbf{y}$.

Then, one proceeds by laying down rules, called *inductive clauses*, by which more complex objects (called *composite objects*) can be constructed from simpler components.

Example. Rules 3–5 were our inductive clauses. The composite objects have the forms $\neg\mathbf{E}$, $\mathbf{E} \wedge \mathbf{F}$, or $(\forall\mathbf{v})\mathbf{E}$.§

When we have to prove that *all* of some type of object (here, set-theoretic propositions) satisfy some predicate (here, "$\mathbf{v}$ and $\mathbf{w}$ are interchangeable within the proposition x"), and those objects are defined inductively, we often use a technique called *proof by structural induction*. Here is how it works.

First, prove that the predicate applies to the objects defined by the basic clauses. This is called the *basic step* of the proof.

*

Strange that the first statement we prove is not a theorem, but a *metatheorem*.

A *theorem* is a proven proposition about sets, or more generally about some mathematical objects. Theorems are stated within a certain mathematical language, and together they form a mathematical theory.

A *metatheorem* is a proven statement *about theorems*, or about the language in which they are spoken, or about the theory they form, rather than about mathematical objects themselves.

In this context, the Latin prefix "-meta" means "beyond." Hence, metatheorems are beyond theorems, in the sense that they step outside a theory and speak about it from above.

Metatheorem 1 (Synonymy) is a metatheorem because it is talking about $\mathbf{v}$ and $\mathbf{w}$, and here $\mathbf{v}$ and $\mathbf{w}$ are variables, which *represent* sets, not sets themselves.

†

This assumption gives us information to use in our proof, while the remaining vagueness (we don't know which specific variables they are) lends generality to our conclusions. That is, we know we're only talking about variables, and our conclusions will apply to *all* variables.

There is a difficulty/reward trade-off here. If you assume a lot, you will have lots of information to work with and it will be easier to prove your claim, but the resulting theorem will only apply to objects satisfying your assumptions. If you assume less, your work will be harder but more broadly applicable.

‡

Rule 1 defines variables, and is separate from the inductive definition of primitive expressions.

Then, for each inductive clause, show that *if* the predicate applies to the components of the clause, it must therefore apply to the resulting composite object. These are called the *inductive steps* of the proof. A proper proof by structural induction requires one inductive step for every inductive clause in the definition of the objects in question. Each inductive step involves proving a conditional statement: *if* the predicate applies to the components, *then* it applies to the construct. The premise of this conditional is called the *inductive hypothesis* for that particular inductive step.

If you can successfully make the basic step and the inductive steps, then you can conclude that the property holds for *all* examples. To see why, let's look at a simpler kind of inductively defined object: the numbers 0, 1, 2, 3, etc. These are called *natural* numbers, and they are inductively defined as follows.

Basic clause: 0 is a natural number.

Inductive clause: if n is a natural number, $S(n)$ is a natural number.

This is a simple inductive definition because there is only one inductive clause. We start with 0. Applying the inductive clause yields $S(0)$, which we name "1". Next we construct $S(1)$ (which is really $S(S(0))$), and name it "2", and so on. You can see that with this simple 2-sentence definition, we have defined infinitely many objects, each built from the previous.

The objects themselves are constructed in sequence, and we can prove things about them following the same sequence. If we want to prove that some predicate P applies to every natural number, we first prove that it applies to 0. Then we show that, if it applies to n, then it applies to $S(n)$ (i.e., $n+1$).

Now, suppose we've done all that: we know that $P(0)$ is true, and also that $P(n) \implies P(n+1)$. Then we can conclude by modus ponens that $P(1)$ is true, since $P(0)$ and $P(0) \implies P(0+1)$. And by the exact same argument we conclude that $P(2)$, and hence $P(3)$, $P(4)$, etc. We can imagine that, as the numbers are defined in the first place, they are hounded by this proof biting at their heels, proving that each newly constructed number satisfies the predicate because the previous one did.

Of course, inductive definitions aren't usually so simple. In the case of our primitive expressions, we have *three* inductive clauses, not just one. Continuing the "hound" analogy, from each component object, we might need to pursue our prey in one of three different directions. Therefore, we need *three* inductive steps in our proof: one for each inductive clause. If we can do this, then our proof can hunt each expression however it turns along its construction.

Enough talk: let's prove it.

Proof. Let $\mathbf{v}$ and $\mathbf{w}$ be variables such that $\mathbf{v} = \mathbf{w}$.*

We first proceed by structural induction to prove that, assuming $\mathbf{v} = \mathbf{w}$, $\mathbf{v}$ and $\mathbf{w}$ are interchangeable within every set-theoretic proposition.†

First note that if a proposition doesn't contain any occurrences of the variable being replaced, then the replacement of the nth occurrence has no effect and therefore obviously the truth value of the proposition doesn't change.

Basic step:‡

1. By Definition 1 (=), we have:

$$(\forall x)(x \in \mathbf{v} \iff x \in \mathbf{w}).$$

§

These forms might seem simple until one remembers that "**E**" might be something like $(\forall x)(\forall y)((x \in y) \wedge \neg(y \in x))$.

*

This "let" phrasing is common in proofs as a way to introduce (here, meta-)variables and what we are assuming about the objects they represent. You will also become very familiar with the phrase "such that".

†

It's important to specify which inductively-defined objects our proof concerns, so that the reader can follow along with the definition.

‡

It's customary in an inductive proof to explicitly label the basic and inductive steps.

That is, $\mathbf{v}$ and $\mathbf{w}$ are interchangeable in the right-hand position.

2. By Axiom 1 (Extensionality):

$$(\forall x)(\mathbf{v} \in x \iff \mathbf{w} \in x).$$

That is, $\mathbf{v}$ and $\mathbf{w}$ are interchangeable in the left-hand position.*

> *
>
> This little point is the sole reason we stated the axiom of extensionality, one of the merely nine assumptions upon which all of mathematics is founded. It will never be mentioned again.

Therefore, $\mathbf{v}$ and $\mathbf{w}$ are interchangeable within all basic propositions of the form "$x \in y$."
Inductive steps: Suppose that $\mathbf{v}$ and $\mathbf{w}$ are interchangeable within the propositions $\mathbf{P}$ and $\mathbf{Q}$. Then:

1. Substituting any occurrence of $\mathbf{v}$ for $\mathbf{w}$ within $\neg\mathbf{P}(\mathbf{v})$ only requires making the change within $\mathbf{P}(\mathbf{v})$. And by our inductive hypothesis, $\mathbf{P}(\mathbf{v}) \iff \mathbf{P}(\mathbf{w})$. Hence $\neg\mathbf{P}(\mathbf{v}) \iff \neg\mathbf{P}(\mathbf{w})$, as a simple fact of logic. The analogous argument applies when substituting $\mathbf{w}$ for $\mathbf{v}$ within $\neg\mathbf{P}(\mathbf{w})$. Therefore, $\mathbf{v}$ and $\mathbf{w}$ are interchangeable within $\neg\mathbf{P}$.

2. For analogous reasons, $\mathbf{v}$ and $\mathbf{w}$ are interchangeable within $\mathbf{P} \wedge \mathbf{Q}$.

3. For $(\forall \mathbf{x})\mathbf{P}$:

   (a) If $\mathbf{x}$ represents the variable being replaced, then there will be no free occurrences of that variable within $\mathbf{P}$ and hence the substitution will have no effect (by the definition of substitution given above).

   (b) If $\mathbf{x}$ represents the replacing variable, then the resulting occurrence would be bound and hence the substitution will have no effect (by definition).

   (c) If $\mathbf{x}$ doesn't represent either of them, then $\mathbf{v}$ and $\mathbf{w}$ are interchangeable as with the previous inductive clauses.

Thus, if $\mathbf{v}$ and $\mathbf{w}$ are interchangeable within $\mathbf{P}$ and $\mathbf{Q}$, then they are interchangeable within every composite expression that can be constructed from $\mathbf{P}$ and $\mathbf{Q}$ by the inductive clauses.

Therefore, by structural induction, $\mathbf{v}$ and $\mathbf{w}$ are interchangeable within every primitive expression. Hence they are interchangeable within every primitive proposition, since every primitive proposition is a primitive expression. Thus, they are interchangeable within *every* set-theoretic proposition, since, while there exist set-theoretic propositions that are not primitive, they can all be easily translated into primitive propositions by replacing defined terms by their definitions. Therefore, $\mathbf{v}$ and $\mathbf{w}$ are synonymous, by definition.

Since we assumed only that $\mathbf{v} = \mathbf{w}$ and found that $\mathbf{v}$ is synonymous with $\mathbf{w}$, we can deduce that:

$$(\mathbf{v} = \mathbf{w}) \implies (\mathbf{v} \text{ is synonymous with } \mathbf{w}).$$

To demonstrate the converse, observe that it is tautological that:

$$(\forall x)(x \in \mathbf{v} \iff x \in \mathbf{v}).$$

And therefore, if $\mathbf{v}$ and $\mathbf{w}$ are synonymous, we can replace the right-hand occurrence of $\mathbf{v}$ with $\mathbf{w}$ to obtain:

$$(\forall x)(x \in \mathbf{v} \iff x \in \mathbf{w}).$$

Which is exactly to say that $\mathbf{v} = \mathbf{w}$. Therefore,

$$(\mathbf{v} \text{ is synonymous with } \mathbf{w}) \implies (\mathbf{v} = \mathbf{w}).$$

Combining this result with the previous one, we conclude the proof: for any two variables $\mathbf{v}$ and $\mathbf{w}$, $(\mathbf{v} = \mathbf{w}) \iff (\mathbf{v} \text{ is synonymous with } \mathbf{w})$.* □

*

When, in the course of a proof, someone writes a little □, or "Q.E.D.", that means the proof is over.

7 Conclusion

Having proven Metatheorem 1 (Synonymy), we have justified our definition of identity. From now on, if we determine that "two sets" have the same elements, we can consider them the same set and interchange one for the other wherever we like. This concludes our discussion of identity, and of the axiom of extensionality. Do Exercise 7.1.

Exercises

7.1 ⓢ blah

Solutions to exercises

*

Admittedly, it is possible for a single word to act as a proposition in certain contexts.

Example. One person says "Cats are reptiles," and another replies "Wrong!" Here the single word "wrong" is used to claim that cats are not reptiles.

Preface

3.1 Congratulations! You have completed the first exercise.

Chapter 1: Mathematics and logic

2.1

1. Does a question describe reality? One might think *no*, but perhaps this is not so obvious. To ask a question does *imply* something about reality: namely, that you want an answer. Yet some questions are rhetorical, or asked to feign ignorance, etc.

   Because English is so vague, we could go either way on this one. But if questions were propositions, it would be difficult to find a sentence that wasn't. Therefore, it is my opinion that questions are *not* propositions.

2. There is some room for disagreement as to what "wet" really means, and whether water should be considered something that *is wet* or rather the substance of wetness itself. However, under either of these interpretations our sentence would describe reality: either water is something wet or it isn't, and either it is the substance of wetness or it is not. Therefore, I would classify it as a proposition, but perhaps an ambiguous one.

3. This is essentially the same as the question about "Who are you?". It is implied that the speaker wants you to close the door, but that's really not how they're expressing themselves. It's not a proposition.

4. Whether $2 + 2 = 1000$, given the usual definitions of these symbols, is a matter of objective, mathematical fact. So yes, this is a proposition, and perhaps our clearest example yet.

5. The word "proposition" is not a proposition, it is just a word.*

6. I'll leave this to your own philosophy.

7. This strange sentence, called the *Liar Sentence*, has historically been the subject of much discussion, some of which we will discuss now, some later, and most never.

   For now, I draw your attention to three properties of the Liar Sentence:

   i. The subject of the Liar Sentence is a sentence.

> *
>
> In the previous example, I was only claiming something about the words contained in the sentence, so we only had to refer to its written form, not its meaning or accuracy. We cannot do that here, because the truth or untruth of a claim cannot be determined just by looking at squiggles on a page, but by interpreting what they mean and comparing the picture thus generated against reality and seeing if they match.

From this we can deduce that it *is* indeed talking about reality, because sentences are part of reality. In and of itself, referring to a sentence is no obstacle to propositionality. For example, I *propose* that the sentence "cats are reptiles" contains the word "hyenas". This claim of mine certainly describes reality, and in fact does so inaccurately.

ii. The Liar Sentence claims that its subject sentence is *not true*.
Again, there are plenty of propositions about the non-truth of sentences: for example, I claim that the sentence "cats are reptiles" is not true. And again, this claim is a description of reality; this time, an accurate one. But when one sentence claims the untruth of another, it becomes dependent upon it. That is, before we can determine the accuracy, or even the meaning, of the denying sentence, we must determine the accuracy or meaning of the sentence it denies.*

iii. The subject sentence which the Liar Sentence refers to is the Liar Sentence itself; that is, the Liar Sentence is self-referential.
Now, there are plenty of meaningful self-referential propositions: for example, "this sentence contains the word 'sentence'." But the Liar Sentence combines self-reference with a claim *of untruth*: therefore, it places itself in a position of self-dependency. That is, before we can determine the meaning of the Liar Sentence, we must *first* determine the meaning of the Liar Sentence.

What does the Liar Sentence mean? Well, depends on what the Liar Sentence means. Okay, so what does it mean? Depends on what it means. Etc.

That is, the Liar Sentence never obtains any meaning: it only makes a claim about itself, and the particular kind of claim it makes is such that it is meaningless until it has already become meaningful.

Therefore no, the Liar Sentence is not a proposition, because while its subject is a part of reality (namely, itself), it doesn't say anything meaningful about that part of reality.

3.1 First, let us recall what we found earlier about the Liar Sentence: it does refer to a portion of reality (namely, itself), but it doesn't say anything meaningful about it. Therefore, it cannot be true, and it also cannot be false, because it does not generate any description which could be accurate or inaccurate of reality. Therefore, the answer is simply: *neither*.

One might be surprised by this result, feeling perhaps that every declarative English sentence should be either true or false: but what about "today, the temperature is cylindrical"? This sentence is perfectly grammatical, but also perfectly meaningless, and is therefore not *truth-apt*: we cannot say it is true, and we cannot say it is false, because it doesn't mean anything.

One might also respond: "The Liar Sentence claims that it is not true. If you've decided that it is neither true nor false, then you must agree that it is not true. (So far, this is correct.) Therefore, you must agree that its self-description is accurate, and therefore true. Etc."

But this makes a subtle mistake: the Liar Sentence *does not* claim that it is not true, for it demonstrably does not claim *anything*. The Liar Sentence *cannot be true*

because, again, it does not generate any description of reality whatsoever (not even of itself).

4.1 $\neg p$ claims that p is false. So, if p *is* indeed false, then $\neg p$'s claim is accurate, so it is true. That is, the truth value of $\neg p$ is $\top$.

Example. Consider the proposition "Rocks are squishy." This is false, and so the sentence "Rocks *aren't* squishy" is true.

4.2 First, $\neg p$ claims that p is false, so $\neg\neg\bot$ claims that $\neg\bot$ is false. Is it? $\neg\bot$ claims that $\bot$ is false, which is true! So $\neg\bot$ is true. Therefore $\neg\neg\bot$'s claim that $\neg\bot$ is false is inaccurate. That is, $\neg\neg\bot$ is false.*

4.3 $p \wedge q$ claims that p and q are *both* true.† So if p is true, that's half the battle: we need q to be true as well. If q is false, then $p \wedge q$ is false: it claims both are true, but only one is true. Therefore, yes, if p is true, $p \wedge q$ can still be false.

4.4 If p is false, then it can't be the case that both p and q are true, so $p \wedge q$ is false. Therefore, no, if p is false then $p \wedge q$ *cannot* be true.‡

4.5 Here I am referring to the Law of Noncontradiction and the Law of the Excluded middle. The latter says that every proposition must be either true or false: there are no other options. The former further specifies that no proposition can be *both* true and false: they are mutually exclusive. Together, the two "laws" state that every proposition is true or false, but not both. Therefore, under these assumptions, it is the *exclusive or* that stands between truth and falsehood.

4.6 $p \vee q$ is a weaker claim than $p \wedge q$:§ it only claims that *one* of p or q is true. That is, it's fine for one to be false, as long as the other is true. Hence, if p is true, then $p \vee q$ is already satisfied, regardless of q. Therefore: no, if p is true, then $p \vee q$ cannot be false.¶

4.7 $p \vee q$ merely claims that one of its operands are true, not both. So, if p is false, there is still hope for q to be true: then $p \vee q$ would be satisfied. Therefore, yes, if p is false, then $p \vee q$ might still be true.‖

4.8 We know that $p \implies q$ is synonymous with $\neg(p \wedge \neg q)$. Therefore, if $p \implies q$ is false, then $\neg(p \wedge \neg q)$ is false, which means $p \wedge \neg q$ is true. Hence p is true and q is false. That is, the truth values of p and q are $\top$ and $\bot$, respectively.

4.9 We're told to assume that $p \implies q$ is true, which is to say that $\neg(p \wedge \neg q)$ is true. That is, we cannot have both p being true and q being false. Therefore, if we know that p is true, then q must be true: otherwise, we would have both p and $\neg q$. Therefore, the truth value of q is $\top$.**

4.10 This is very similar to the last question. If $p \implies q$ is true, that means $\neg(p \wedge \neg q)$ is true. Thus p and $\neg q$ cannot both be true. But we are told that q is false (i.e., $\neg q$ is true), so p must be false: otherwise we would have both p and $\neg q$. Therefore, the truth value of p is $\bot$.††

*

Observe that adding a '$\neg$' toggles the truth value of its operand,[a] from $\top$ to $\bot$ or from $\bot$ to $\top$.

[a] An *operand* is something to which an operator applies. That is, it's the thing (or things) the operator *operates* upon. E.g., in $\neg p$, the operand of the denial is p.

†

Are you noticing a pattern in these solutions, yet?

‡

Notice that you don't even need to know what q *is*: if you already know p is false, you can immediately infer that $p \wedge q$ is false as well.

§

When I say a *weaker* claim, I mean one that doesn't claim as many things.[a] Symmetrically, a *stronger* claim is one that claims more things.[b]

[a] This terminology is perhaps misleading. In a sense, claiming fewer things makes a statement more resilient: there are fewer ways to disprove it.

[b] And is therefore harder to prove and easier to disprove.

¶

Notice the symmetry between this result and the previous result about conjunction: if p is false, we can short-cut and conclude that $p \wedge q$ is false, and if p is true, we can short-cut and conclude that $p \vee q$ is true.

4.11 As we've seen in previous solutions, $p \implies q$ claims that p and $\neg q$ are not both true. That is, it claims that either p is false or q is true (or both). If the premise (p) is false, then clearly this claim is accurate. Therefore, no, if the premise of a conditional is false then that conditional cannot be false.*

4.12 Very similar to the last question. If q is true, we have already satisfied the claim made by $p \implies q$: that p is false or q is true. Equivalently, we know that for $p \implies q$ to be false we would need p to be true and q to be false: if q is true, this cannot be the case. Therefore, no, if the conclusion of a conditional is true, then that conditional cannot be false.

4.13 In Exercise 4.9 we found that, when we already know that $p \implies q$ is true, then, if p is true, we can deduce q. That is, we can *conclude* q.

4.14 There are two reasons.

1. As we have seen, $p \implies q$ can only be false if p is true. Therefore, p serves as a *condition* for the falsifiability of $p \implies q$.

   But if a proposition cannot be falsified, then it makes no claim about reality.† Hence, a conditional really only speaks about the circumstance in which its condition is met (i.e., is true): if the condition is false, it says nothing. Therefore, p serves also as a qualification for its conditional: it restricts the scope of its claim. When the condition is met, our circumstance lies within the jurisdiction of the conditional and its claim applies.

2. Suppose we already know that $p \implies q$, and we are trying to prove q.‡ As we have seen in previous exercises, because we know $p \implies q$, we can deduce the conclusion q as soon as we determine that the condition, p, is true. Therefore, when $p \implies q$ is true, p serves as a condition for the truth of q.

In summary, we call "$\implies$" the "conditional" operator because it produces propositions that are qualified by a condition: that is, statements that only assert their claim when their condition is met.

4.15 As we have seen in a previous exercise, if "$p \implies q$" is true, then, if we know that p is also true, we can deduce q. Therefore it is natural to pronounce "$p \implies q$" as if it were an English *if... then* statement: "if p, then q."

However, there is an important difference. To see it, consider the example "The sun is hot $\implies$ these words are written on a page." The sun is indeed hot, and these words are indeed written on a page. Therefore the conditional is true. But in ordinary English, it seems strange to say that "if the sun is hot, then these words are written on a page," because these two facts have nothing whatsoever to do with one another.

When we make an *if... then* statement in English, we require that the premise is somehow *relevant* to the conclusion: specifically, the premise should *necessitate* the conclusion. That is, it should not only be *actually not the case* that the premise is true and the conclusion false, but this state of fairs should be *impossible*. As an English *if... then* statement, "if the sun is hot, then these words are written on a page" is false, because we can easily imagine a world in which the sun is hot but these words are spoken rather than written on a page, even though this is not actually the case.

‖

This can also be seen from the perspective of the previous exercise. If q is true, it doesn't matter what p is: $p \vee q$ will be true regardless.

**

We will use this particular deduction constantly in mathematics. It is known as the rule of *modus ponens*.[a]

[a]From Latin, meaning "the method of affirming": in this context, the method of *concluding*.

††

Like *modus ponens*, this method of deduction is used frequently in mathematical reasoning. It is called *modus tollens*.[a]

[a]From Latin, meaning "the method of taking away": in this context, the method of *denying*.

*

Another equivalent way of seeing this is that, as we found in the solution to a previous exercise, for $p \implies q$ to be false we would need p to be true and q to be false. Since p is false, clearly that isn't the case here.

†

If it did, then that claim could be falsified by reality.

‡

As you will soon see, mathematicians are in this situation regularly.

To answer the question, the relation between a logical conditional and an English *if... then* statement is:

1. A logical conditional merely claims that it is not actually the case that its premise is true and its conclusion false.
2. An English *if... then* statement claims that it is impossible for its premise to be true and its conclusion false. In other words, it claims that the corresponding logical conditional is *necessary.**

We will be able to understand this more precisely having been introduced to certain concepts later in this chapter, and at that time I will give you a better explanation.

*

In philosophy, by a "necessary" proposition we mean one that *must* be true. That is, one that cannot be false.

†

For the same reason, we use " $\iff$ " to combine the arrows from $p \implies q$ and $q \implies p$, the latter of which could alternatively be written as $p \impliedby q$.

4.16 In general, there are exactly four possible combinations of truth values for two propositions p and q.

1. Both are true.
2. p is true, but q is false.
3. p is false, but q is true.
4. Both are false.

If $p \implies q$, then we cannot have "p is true, but q is false." If, in addition, $q \implies p$, then we can further rule out "q is true, but p is false." The only remaining possibilities, having removed these two, are:

1. Both are true.
2. Both are false.

That is, p and q have the same truth value.

4.17 In Exercise 4.16, we found that, if $p \implies q$ and $q \implies p$, then p and q have the same truth value. But this is exactly what $p \iff q$ means. Therefore we name this operator the "biconditional", after the two (i.e., <u>*bi*</u>) conditionals $p \implies q$ and $q \implies p$.†

4.18 First, recall that a biconditional is a conjunction of two conditionals: "$p \iff q$" is synonymous with "$(p \implies q) \wedge (q \implies p)$." So, when we say "if and only if," the "if" corresponds to one of the two conditionals, and the "only if" corresponds to the other, with a "the" in-between for the conjunction.

In English, "p if q" is means the same thing as "if q, then p." Therefore, the "if" corresponds to $q \implies p$.

To say that "p only if q" would be to claim that p cannot be true unless q is also true: that is, that it is not the case that p is true and q is false. But this is exactly what is claimed by $p \implies q$. Therefore, the "only if" corresponds to $p \implies q$.

4.19 $p \iff q$ claims that p and q have the same truth value. Therefore, it is false when their truth values differ: that is, when $p \wedge \neg q$ or when $\neg p \wedge q$.

*

The order in which we write the combinations is arbitrary, but it is useful to have a consistent convention. The most widely accepted convention, and the one we will use in this book, is to treat $\bot$ as 0 and $\top$ as 1 and count upward from 0 to $2^n - 1$ in binary (i.e., base 2).

Example. To determine the order of combinations of truth values for the conjunction table, which has 2 variables and therefore $2^2 = 4$ combinations of truth values, we count up from 0 to 3 in binary. The first digit of the binary representation corresponds to the truth value of p, and the second digit corresponds to the truth value of q.

Count	Binary	p	q
0	00	$\bot$	$\bot$
1	01	$\bot$	$\top$
2	10	$\top$	$\bot$
3	11	$\top$	$\top$

4.20 Since $p \iff q$, we know p and q must have the same truth value. Therefore, since p is true, we deduce that q must also be true.

4.24 According to the Law of the Excluded Middle, p must be either true or false. That is, either p is true or $\neg p$ is true. In either case, $p \vee \neg p$ is true. So no, $p \vee \neg p$ cannot be false.

4.25 If p is false and q is true, then neither p nor $\neg q$ are true, and therefore $p \vee \neg q$ is false. So yes, $p \vee \neg q$ can be false, depending on the truth values of p and q.

5.1 For each individual propositional variable (e.g., p or q), there are two possible truth values: $\top$ and $\bot$. Therefore, since our formula (i.e., $p \wedge q$) involves two variables, there are $2 \times 2 = 4$ combinations of truth values. For a formula involving three variables, there would be $2 \times 2 \times 2 = 8$ combinations of truth values. In general, for a formula involving n variables, there are 2^n combinations of truth values.

5.2 First, we write down all combinations of truth values for p and q.*

p	q
$\bot$	$\bot$
$\bot$	$\top$
$\top$	$\bot$
$\top$	$\top$

Next, we add a column for $p \vee q$ and finish drawing the frame of the table.

p	q	$p \vee q$
$\bot$	$\bot$	
$\bot$	$\top$	
$\top$	$\bot$	
$\top$	$\top$	

Now, for each combination (i.e., row), we determine the corresponding truth value of the disjunction.

1. When p is false and q is false, $p \vee q$ is false. That is, $\bot \vee \bot \iff \bot$.
2. $\bot \vee \top \iff \top$.
3. $\top \vee \bot \iff \top$.
4. $\top \vee \top \iff \top$.

Finally, we fill in the column for $p \vee q$ with these truth values.

p	q	$p \vee q$
$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$
$\top$	$\bot$	$\top$
$\top$	$\top$	$\top$

5.3 This time, only one variable is involved: p. Therefore there are only $1 \times 2 = 2$ combinations of truth values.

p
$\bot$
$\top$

And here is the resulting table.

p	$\neg p$
$\bot$	$\top$
$\top$	$\bot$

5.4 The truth table for the conditional is as follows.

p	q	$p \implies q$
$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\top$
$\top$	$\bot$	$\bot$
$\top$	$\top$	$\top$

Let me explain the note I attached to this exercise. We could have done this by simply adding a column to our disjunction table, like so.

p	q	$p \vee q$	$p \implies q$
$\bot$	$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\top$	$\bot$
$\top$	$\top$	$\top$	$\top$

This is (only) possible because disjunction and the conditional involve the same number of variables, and therefore their tables have the same number of rows. We could *not* do this with denial, because our denial table only had two rows, and the disjunction table requires four.*

5.5 The truth table for the biconditional is as follows.

p	q	$p \iff q$
$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\bot$
$\top$	$\bot$	$\bot$
$\top$	$\top$	$\top$

5.6 For illustration, I will set up a single truth table showing the columns for both the biconditional and the conjunction of conditionals.

*

However, we *could* do it the other way around: that is, we could construct a table for denial by adding a column to our conditional table, like so.

p	q	$p \implies q$	$\neg p$
$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\bot$
$\top$	$\top$	$\top$	$\bot$

Observe that rows 2 and 4 are redundant for the purposes of the denial column, because they have the same truth value for p and our denial doesn't involve q.

*

Notice that this column is *not* the same as the column we obtained for the original equivalence: this deductive process doesn't produce a formula equivalent to the original, but *two* formulas equivalent to each other (but not necessarily the original).

p	q	$p \iff q$	$(p \implies q) \wedge (q \implies p)$
$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\bot$	$\bot$
$\top$	$\bot$	$\bot$	$\bot$
$\top$	$\top$	$\top$	$\top$

As you can see, the two columns are identical. That is, for any given combination of truth values of their variables, the two formulas have the same truth value. This is comforting, given that we previously found them to be synonymous.

6.1 Here is the truth table, with intermediary columns omitted.

p	q	$\neg p \implies (p \vee q)$	$\neg(\neg p \wedge \neg(p \vee q))$
$\bot$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\top$	$\top$
$\top$	$\top$	$\top$	$\top$

The columns are the same, so the equivalence is indeed true.*

6.2 The truth table is as follows.

p	q	$(p \implies q) \wedge (q \implies p)$	$\neg(p \wedge \neg q) \wedge (q \implies p)$
$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\bot$	$\bot$
$\top$	$\bot$	$\bot$	$\bot$
$\top$	$\top$	$\top$	$\top$

Because the columns are the same, we find that the equivalence is true.

6.4 Claiming "$p \wedge p$" is essentially claiming p twice.

Examples. Here are some propositions with this form.

1. I live in Canada, and I live in Canada.
2. The moon is made of cheese, and the moon is made of cheese.

So "$p \wedge p \equiv p$" states that making a claim twice doesn't change whether it is true or false.

Let's see if that's accurate. The truth table for $p \wedge p$ is as follows.

p	$p \wedge p$
$\bot$	$\bot$
$\top$	$\top$

Since the columns for p and for $p \wedge p$ are identical, we conclude that the logical equivalence $p \wedge p \equiv p$ is true.

There is a reason why, when someone says the same thing twice, we call their communication "redundant" and consider the second statement unnecessary: it doesn't (explicitly*) communicate any additional information. That is, no more (and no less) is claimed by saying something twice rather than once. Therefore, there is no way for the truth value to change. If you make an incorrect statement twice, you're still wrong. And if you make a correct statement twice, you are being redundant but are correct nonetheless.

> *
>
> In natural language, repetition can implicitly carry additional information, such as emphasis, emotion, etc. When stating mathematical propositions in the language of logic, we tend not to do this.

6.5 This is similar to the previous exercise, but with disjunction, rather than conjunction, standing between p and p. A proposition of the form "$p \vee p$" claims that either p is true, or p is true, or both. Our proposed equivalence, "$p \vee p \equiv p$", states that this is the same as simply claiming p.

To test this, we will build a truth table.

p	$p \vee p$
$\bot$	$\bot$
$\top$	$\top$

From this table, we can see that the two formulas have the same truth value in every possible circumstance. Therefore, the proposed equivalence is true.

Here is how I make sense of it. Ordinarily, a disjunction claims of two propositions that at least one is true. Such a claim is accurate if one is true and the other false, or if both are true, and is falsified only if both are false.

But in our case, the two claims are the same: they are both p. Hence it is impossible for one to be true and "the other" false. Therefore, in our circumstance, the disjunction is true only when both p and p are true. But this is exactly when $p \wedge p$ is true, so we would expect that $p \vee p \equiv p \wedge p$. Indeed this is true, as you may verify with a truth table. And as we learned in the previous exercise, $p \wedge p \equiv p$. Therefore, we can deduce that $p \vee p \equiv p$, by subformula replacement.

6.6 The proposition $\neg\neg p$ is a *double denial* of p: it claims that "it is not the case that it is not the case that p," or more simply that "p is not *not* true," or even more concisely, that "p is not false." Thus, the proposed equivalence, $\neg\neg p \equiv p$, states that to claim that "p is not false" is to claim that "p is true."

Let's make a truth table, using an intermediary column for $\neg p$ to make it more clear what's going on.

p	$\neg p$	$\neg\neg p$
$\bot$	$\top$	$\bot$
$\top$	$\bot$	$\top$

Comparing the columns for p and $\neg\neg p$, we can see that they are indeed equivalent.

As we discussed previously, the denial operator flips the truth value of its operand: either from $\top$ to $\bot$, or from $\bot$ to $\top$. So, when we apply the first denial, we flip the truth value. But, when we apply the second denial, we flip it back again and end up where we started: with p.

*

Here I show a chain of equivalences, indicating that every formula in the chain is equivalent to the others: if $A \equiv B$ and $B \equiv C$, then $A \equiv C$ (because one can replace B with C within $A \equiv B$ by subformula replacement).

6.7 This equivalence states that to claim that "p is not *not* false" is to claim that "p is true."

Before we make a truth table, let's take a shortcut and use deduction. We already know that $\neg\neg p \equiv p$. Our proposition "$\neg\neg\neg p$" is a substitution instance of this formula, by the replacement of p with $\neg p$. Therefore we deduce the equivalence $\neg\neg\neg p \equiv \neg p$. That is, to say that a propositions isn't *not* false is to say that it *is* false. Since our original equivalence claimed the opposite, that original equivalence was incorrect.

Let's verify this with a truth table.

p	$\neg p$	$\neg\neg p$	$\neg\neg\neg p$
$\bot$	$\top$	$\bot$	$\top$
$\top$	$\bot$	$\top$	$\bot$

We see that the columns for p and $\neg\neg\neg p$ are different, and therefore the equivalence between them is false. Instead, the column for $\neg\neg\neg p$ matches that of $\neg p$: the exact opposite of p.

As we found in the last exercise, after two denial (i.e., $\neg\neg p$) we have arrived back at the truth value of p. After denying once more, we reach $\neg p$.

6.8 Having completed the previous two exercises, we should expect this equivalence to be true. By deduction:

$$\begin{aligned}\neg\neg(\neg\neg p) &\equiv \neg\neg p \\ &\equiv p.^*\end{aligned}$$

Of course, we can verify this with a truth table.

p	$\neg\neg\neg p$	$\neg\neg\neg\neg p$
$\bot$	$\top$	$\bot$
$\top$	$\bot$	$\top$

In general, an even number of denial completely cancels out, and an odd number reduces to a single denial.

6.9 This equivalence claims that the truth value of a conjunction isn't affected by the order of the two operands.

We don't have any relevant equivalences for the use of deduction, so we will check this with a truth table.

p	q	$p \wedge q$	$q \wedge p$
$\bot$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\bot$	$\bot$
$\top$	$\bot$	$\bot$	$\bot$
$\top$	$\top$	$\top$	$\top$

So yes, this equivalence is true: the order of operands doesn't affect the truth value of a conjunction. This property is called *commutativity*, and we say that conjunction "commutes" or is "commutative." "Com-" is a Latin prefix meaning "with" or "together," and to "-mutate" is to change. Therefore commutativity is so named for "changing with another": in this case, swapping places with another. When we encounter a new mathematical operator, we will always be interested in whether it commutes, as it is useful to be able to swap the order of things.

It makes sense that conjunction would be commutative, because a conjunction claims that its operands are both true, which is a symmetric claim: no mention is made of a "first" or "second" operand.

*

This equivalence is called *De Morgan's first law*, named after the mathematician and logician Augustus De Morgan. Soon we will discover his second law.

6.10 A proposition of the form $\neg(p \wedge q)$ claims that it isn't the case that p and q are both true: that is, it claims that they aren't both true.

A proposition of the form $\neg p \wedge \neg q$ claims that p is false and that q is false: that is, it claims they are both false.

Therefore, the equivalence $\neg(p \wedge q) \equiv \neg p \wedge \neg q$ states that "to say that p and q aren't both true is to say they are both false."

Here's the truth table.

p	q	$\neg(p \wedge q)$	$\neg p \wedge \neg q$
$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\top$	$\bot$
$\top$	$\bot$	$\top$	$\bot$
$\top$	$\top$	$\bot$	$\bot$

These columns are not the same, so the equivalence is false.

Just because p and q aren't both true doesn't mean they're both false. Look at the truth table: when only one of p and q are true and the other is false, it is accurate to say that they aren't both true (because one is false), but inaccurate to say they are both false (because one is true).

6.11 This equivalence states: to say that p and q aren't both true is to say that one or both are false.

Here is the truth table.

p	q	$\neg(p \wedge q)$	$\neg p \vee \neg q$
$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\top$	$\top$
$\top$	$\top$	$\bot$	$\bot$

Indeed, they are equivalent.

This makes sense: if they aren't both true, then one must be false (otherwise they would both be true). Conversely, if one is false, then they can't both be true.*

*

This equivalence is the aforementioned *De Morgan's second law*.

6.12 This equivalence states that disjunction is commutative.

Let us construct the relevant table.

p	q	$p \vee q$	$q \vee p$
$\bot$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\top$	$\top$
$\top$	$\top$	$\top$	$\top$

So yes, disjunction is commutative.

This makes sense, because like a conjunction, a disjunction makes a symmetric claim: "at least one of my operands are true."

6.13 A proposition of the form $\neg(p \vee q)$ claims that it is not the case that at least one of p and q are true: that is, that neither of them are true.

As discussed previously, a proposition of the form $\neg p \wedge \neg q$ claims that both p and q are false.

Therefore, this equivalence, $\neg(p \vee q) \equiv \neg p \wedge \neg q$, states that to say that neither of p and q are true is to say that they are both false.

Here is the truth table.

p	q	$\neg(p \vee q)$	$\neg p \wedge \neg q$
$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\bot$	$\bot$
$\top$	$\bot$	$\bot$	$\bot$
$\top$	$\top$	$\bot$	$\bot$

The columns match, so the equivalence holds.

This makes sense: if neither p nor q is true, then they must both be false (otherwise one would be true). And conversely, if they are both false, then neither are true.*

6.14 The equivalence $\neg(p \vee q)$ states: "to say that neither p nor q is true is to say that one of them is false."

We consult the truth table:

p	q	$\neg(p \vee q)$	$\neg p \vee \neg q$
$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\bot$	$\top$
$\top$	$\bot$	$\bot$	$\top$
$\top$	$\top$	$\bot$	$\bot$

As you can see, the equivalence is false.

To say that neither p nor q is true is not merely to say that one is false, but that *both* are false: otherwise, one would be true and it wouldn't be accurate to say that neither is true.

6.15 This equivalence states that to say p and q are both true is to say that neither of them are false.

This is a good opportunity to use deduction.

$$\begin{aligned} \neg(\neg p \vee \neg q) &\equiv \neg\neg p \wedge \neg\neg q && \text{by De Morgan's second law.} \\ &\equiv p \wedge q && \text{by double denial.} \end{aligned}$$

Therefore:

$$p \wedge q \equiv \neg(\neg p \vee \neg q).$$

You may verify this with a truth table if you want, but do not feel obligated.

Even in English, it makes sense. If p and q are both true, then neither of them are false: otherwise, they could not both be true. Conversely, if neither of them are false, then they must both be true: for there is no other option besides truth and falsehood.*

6.16 This equivalence states that to say that at least one of p and q are true is to say that they are not both false.

Again, we won't bother with a truth table but deduce the truth directly.

$$\begin{aligned} \neg(\neg p \wedge \neg q) &\equiv \neg\neg p \vee \neg\neg q && \text{by De Morgan's first law.} \\ &\equiv p \vee q && \text{by double denial.} \end{aligned}$$

Thus, the equivalence is true:

$$p \vee q \equiv \neg(\neg p \wedge \neg q).$$

That is, to say that at least one of p and q is true is to say they are not both false (otherwise, *zero* of them would be true, and zero is not at least one).†

6.17 This equivalence states that, when claiming that three things are true, it doesn't matter where you put the parentheses.

To test this, we will require an unusually large truth table.‡

p	q	r	$q \wedge r$	$p \wedge (q \wedge r)$	$p \wedge q$	$(p \wedge q) \wedge r$
$\bot$	$\bot$	$\bot$	$\bot$	$\bot$	$\bot$	$\bot$
$\bot$	$\bot$	$\top$	$\bot$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\bot$	$\bot$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\top$	$\bot$	$\bot$	$\bot$
$\top$	$\bot$	$\bot$	$\bot$	$\bot$	$\bot$	$\bot$
$\top$	$\bot$	$\top$	$\bot$	$\bot$	$\bot$	$\bot$
$\top$	$\top$	$\bot$	$\bot$	$\bot$	$\top$	$\bot$
$\top$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$

*

This equivalence serves as an alternate *definition* for conjunction in terms of denial and disjunction.

†

Similar to the previous equivalence, this one serves as a definition of disjunction in terms of denial and conjunction.

‡

Unfortunately, the next several equivalences involve three variables, which means we need eight rows per table.

Comparing the column for $p \wedge (q \wedge r)$ to that for $(p \wedge q) \wedge r$, we find that they are indeed logically equivalent. That is, in a conjunction, parentheses don't matter.

This property is called *associativity*, because the parentheses associate (i.e., group) things together. We say that conjunction is *associative*. Like commutativity, this is a very useful property to determine, because it lets us ignore parentheses altogether. That is, instead of writing $p \wedge (q \wedge r)$ we just write $p \wedge q \wedge r$.

It's hard to provide an intuitive explanation for why conjunction is associative at this stage, other than to simply say that there is no reason why it wouldn't be. You haven't seen any examples, yet, of operators that are commutative but not associative. Generally, such operators lack associativity because, for whatever reason, the decision of which two operands are operated upon first affects the end result. In the case of conjunction, it doesn't matter.

*

This property is called *distributivity*, and we say that conjunction *distributes* over disjunction, because the "$p\wedge$" seems to visually "distribute" over q and r. This is the same relationship that holds between multiplication and addition: that is, multiplication distributes over addition. As you may know from your experience with arithmetic, this relationship is very useful to establish.

6.18 This equivalence claims that disjunction is associative. I will not waste paper going through all the details, because they are practically identical to those of the previous exercise. Suffice it to say that yes, disjunction is associative.

6.19 A proposition of the form $p \wedge (q \vee r)$ claims that p is true, and that at least one of q and r are true.

A proposition of the form $(p \wedge q) \vee (p \wedge r)$ claims that either p and q are both true, or p and r are both true, or both.

Therefore, the equivalence between these formulas claims that to say that "p is true and at least one of q and r is true" is to say that "either p and q are both true, or q and r are both true (or both)."

Let's check the truth table.

p	q	r	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
$\bot$	$\bot$	$\bot$	$\bot$	$\bot$
$\bot$	$\bot$	$\top$	$\bot$	$\bot$
$\bot$	$\top$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\bot$	$\bot$
$\top$	$\bot$	$\bot$	$\bot$	$\bot$
$\top$	$\bot$	$\top$	$\top$	$\top$
$\top$	$\top$	$\bot$	$\top$	$\top$
$\top$	$\top$	$\top$	$\top$	$\top$

Lo, and behold: it is true.

To understand this intuitively, consider the first formula: $p \wedge (q \vee r)$. For a proposition of this form to be true, p must be true and we must also choose one of q or r (or both) to be true. These two options (q and r) correspond to the two options in the second formula: $p \wedge q$ and $p \wedge r$. The only complication is that we add "$p\wedge$" to the start of each option, to capture the fact that p must be true in either case.*

6.20 Notice the similarity with the last equivalence: the left-hand formula is identical, the right lists two "options" as before, and the first of those options $(p \wedge q)$ is the same. The only difference is in the second option on the right-hand: instead of $p \wedge r$, we have

simply r. Therefore, this equivalence claims that, if you choose r, you don't need p. We expect this to be false, because we have already constructed the truth table for the left-hand formula and it required p to be true in all cases.

Let's test this hypothesis.

p	q	r	$p \wedge (q \vee r)$	$(p \wedge q) \vee r$
$\bot$	$\bot$	$\bot$	$\bot$	$\bot$
$\bot$	$\bot$	$\top$	$\bot$	$\bot$
$\bot$	$\top$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\bot$	$\bot$
$\top$	$\bot$	$\bot$	$\bot$	$\bot$
$\top$	$\bot$	$\top$	$\top$	$\top$
$\top$	$\top$	$\bot$	$\top$	$\top$
$\top$	$\top$	$\top$	$\top$	$\top$

Surprisingly, our intuition was incorrect: the equivalence is true. In a sense, it shows that disjunction and conjunction are "co-associative," which makes sense because, as we have seen before, disjunction and conjunction are both associative separately.

6.21 This equivalence is very similar to the last, differing only in that we have swapped the roles of conjunction and disjunction. In fact, the left-hand formula of this equivalence is almost identical to the right-hand formula of the previous one: the only differences are in the names of the variables and the order in which the operands are written. The same relation holds between the current right-hand formula and the previous left-hand one.

Let's use deduction to re-order things in our equivalence, and rename the variables, to bring it closer to the previous one.* That way, we can more clearly see the relationship between them, without these misleading differences in appearance.

1. First we simply swap the order of the two formulas in the equivalence, justifying this with the commutativity of logical equivalence.† That is, from $p \vee (q \wedge r) \equiv (p \vee q) \wedge r$ we deduce $(p \vee q) \wedge r \equiv p \vee (q \wedge r)$.
2. Next, we swap the order of the operands to the conjunctions in each formula, deducing $r \wedge (p \vee q) \equiv p \vee (r \wedge q)$ by the commutativity of conjunction.
3. Now we do the same with the operands to the disjunctions, deducing $r \wedge (q \vee p) \equiv (r \wedge q) \vee p$ by the commutativity of disjunction.
4. Finally, we rename the variables, which can be thought of as a series of variable expansions.‡ Specifically, we will rename r to p and p to r, leaving q as-is. Thereby, we finally deduce:

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee r.$$

This is exactly the equivalence from the previous exercise. That is, if this equivalence were true, then the previous one would be true. Furthermore, we could simply reverse the deductions we made to obtain the previous equivalence from this one. Therefore

*

We will be deducing the consequences of this equivalence without knowing whether that equivalence is true. This is fine, as long as we are self-aware about it.

†

The commutativity of logical equivalence (that is, the fact that $A \equiv B$ means the same thing as $B \equiv A$) cannot be established with a truth table, because logical equivalence is not a logical operator.[a] But it should be obvious: to say that A and B have the same truth value is to say that B and A have the same truth value. The claim is symmetric.

[a] That is, it does not operate upon atomic propositions to yield composite propositions. Instead, it operates upon formulas to yield atomic propositions.

‡

In this case, the formula isn't really *expanding*, in the sense of getting larger, because the formulas with which we replace its variables are single variables themselves.

the converse also holds: if the previous equivalence is true, then this one is true as well. In other words, they are synonymous.

I will leave the conclusion to you: if the previous equivalence was false, then this one is false too, by modus tollens (see the solution to Exercise 4.10).* If it was true, then this one is true, because they are synonymous.

You may wonder why we bothered with all that, when we could have simply written out the truth table. There are a few reasons:

1. This way is more interesting, and uses more of the deductive techniques we've learned. That is, it is more instructive.
2. Once you are practiced with deduction and it becomes second-nature, it will be much faster than making truth tables, and more integrated with your intuition.
3. When we deal with actual mathematical propositions, rather than formulas and logical equivalences, truth tables will no longer be an option.

6.22 Here the left-hand formula is identical to that of the previous equivalence. However, the right-hand side is different, and together they form an equivalence similar to equivalence 16, which was true. In fact, this equivalence states precisely that disjunction distributes over conjunction.

Here is the truth table.

p	q	r	$p \vee (q \wedge r)$	$(p \vee q) \wedge (p \vee r)$
$\bot$	$\bot$	$\bot$	$\bot$	$\bot$
$\bot$	$\bot$	$\top$	$\bot$	$\bot$
$\bot$	$\top$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\top$	$\top$
$\top$	$\bot$	$\top$	$\top$	$\top$
$\top$	$\top$	$\bot$	$\top$	$\top$
$\top$	$\top$	$\top$	$\top$	$\top$

We find that yes, disjunction distributes over conjunction. To see this more intuitively, let's think about the meaning of each formula.

A proposition of the form $p \vee (q \wedge r)$ asks us to choose at least one of two options: p, and $q \wedge r$. That is, its claim will be satisfied if p is true (regardless of q and r) or if both q and r are true.

A proposition of the form $(p \vee q) \wedge (p \vee r)$ tells us to make two different decisions: between p and q, and between p and r. Notice that in both disjunctions, p is one of the two options. Therefore, if we take p to be true, we have already satisfied both disjunctions. But if we do *not* choose p (that is, if we choose p to be false), then we are left with only one remaining option for each disjunction: q for the first, and r for the second. That is, we can choose p, or, if we do not choose p, we must choose both q and r. This is exactly what the left-hand proposition says, so it makes sense that they are equivalent.†

*

Here we have exemplified the proof technique called *proof by contradiction*. That is, to prove that our equivalence was false, we *assumed that it was true* and deduced various results. One of those results contradicted something we already knew, allowing us to falsify our original assumption via modus tollens. We will prove many, many facts by contradiction.

†

We have now found several connections between conjunction and disjunction.

1. They are both commutative.
2. They are both associative.
3. By De Morgan's first law, the denial of a conjunction is a disjunction.
4. By De Morgan's second law, the denial of a disjunction is a conjunction.
5. Conjunction distributes over disjunction, and disjunction distributes over conjunction.

This is partly why people chose the name "disjunction": because it shares the root "-junction" with "conjunction," but evokes the idea of separation—as in two separate, alternate options, at least one of which must be true.

6.23 When building intuition regarding the conditional, it's useful to think of it as an English *if... then* statement. Although there is an important distinction, which we have discussed previously, that distinction is really only relevant for this chapter, and will practically disappear thereafter throughout our study of mathematics.

Under that interpretation, this equivalence states that to say that p necessitates q is to say that either p is false or q is true (or both).

We will verify this using deduction.

$$\begin{aligned} p \implies q &\equiv \neg(p \wedge \neg q) && \text{by definition.} \\ &\equiv \neg p \vee \neg\neg q && \text{by De Morgan's first law.} \\ &\equiv \neg p \vee q && \text{by double denial.} \end{aligned}$$

So yes, this is true, and could serve as an alternate definition of the conditional in terms of disjunction rather than conjunction.

To understand this intuitively, it is useful to reflect for a moment on the disjunction operator. In my opinion, it is the least intuitive of the five operators we've discussed.

1. A denial claims that something is false.
2. A conjunction claims two things are both true.
3. A conditional claims that one thing leads to another.
4. A biconditional claims that two things each lead to each other.

In the case of each of these four operators, it is obvious to me how such a claim could be true. But a disjunction claims that one of two things must be true, but it doesn't care which one. How could such a fact be enforced by reality?

The answer is to be found in the following equivalence, which might serve as a definition for disjunction:

$$p \vee q \equiv \neg p \implies q.$$

That is, to say that one (or both) of p and q are true is to say that, if p *isn't* true, then q must be. Let's prove this.

$$\begin{aligned} \neg p \implies q &\equiv \neg(\neg p \wedge \neg q) && \text{by definition.} \\ &\equiv p \vee q && \text{as found previously.} \end{aligned}$$

This is how a true disjunction is "enforced" by reality—how it is guaranteed that at least one of p and q are true—if p isn't true, then q is true.

When we view disjunction as a conditional in disguise, our original equivalence becomes more intuitive. To say that $\neg p \vee q$ (that is, to say that either p is false or q is true) is to say that $\neg\neg p \implies q$ (that is, to say that, if p isn't false, q must be true), or more simply that $p \implies q$.

6.24 When you swap the order of a conditional, the result is called the *converse* of the original.*

*

When I use the word "conversely," this is what I mean.

*

Here I'm playing with the tenses so that the English makes sense.

Example. The converse of the conditional "If it rains, then the grass will get wet" is the conditional "If the grass gets wet, it must have rained."*

Therefore, this equivalence claims that a conditional is the same as its converse, which is to say that the conditional operator is commutative.

Here is the truth table.

p	q	$p \implies q$	$q \implies p$
$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\top$	$\bot$
$\top$	$\bot$	$\bot$	$\top$
$\top$	$\top$	$\top$	$\top$

The equivalence is false: the conditional is not commutative, which is to say that a conditional is distinct from its converse.

This makes sense, because a conditional's claim is not symmetric. Consider the equivalence with which we originally defined the conditional: $p \implies q \equiv \neg(p \wedge \neg q)$. This is clearly not symmetric: q is denied, and p is not.

A single example is sufficient to make obvious the noncommutativity of the conditional: "If it rains, the grass will get wet." This is pretty true. However, it is *not* true that "If the grass gets wet, it must have rained," because there are other ways than rain (such as dew) for grass to get wet.

6.25 When you deny the operands to a conditional but don't swap their order, the result is called the *inverse* of the original.

Example. The inverse of the conditional "If it rains, then the grass will get wet" is the conditional "If it doesn't rain, then the grass will not get wet."

This equivalence therefore claims that a conditional is the same as its inverse.

The truth table is as follows.

p	q	$p \implies q$	$\neg p \implies \neg q$
$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\top$	$\bot$
$\top$	$\bot$	$\bot$	$\top$
$\top$	$\top$	$\top$	$\top$

They are not the same, so the equivalence is false: a conditional and its inverse are distinct. Notice also that the column for $\neg p \implies \neg q$ is the same as that for $q \implies p$ in the previous equivalence. Therefore, $q \implies p \equiv \neg p \implies \neg q$, which is to say that a conditional's converse is equivalent to its inverse.

Our rain example clarifies the equivalence between the converse and inverse, as well as the distinction between the inverse and the original conditional. In the previous exercise, we made sense of the difference between a conditional and its converse by noting that there are ways for the grass to get wet that do not involve rain, and that therefore we shouldn't say "If the grass is wet, it must have rained." But this is exactly

why we cannot say that "If it doesn't rain, then the grass will not get wet": there are ways for the grass to get wet, so even if it doesn't rain, the grass might get wet.

6.26 When you deny the operands of a conditional *and* swap their order, the result is called the *contrapositive* of the original.

Example. The contrapositive of the conditional "If it rains, then the grass will get wet" is the conditional "If the grass doesn't get wet, then it must not have rained."

Notice that this is exactly the relationship between $q \implies p$ and $\neg p \implies \neg q$, and we have already found these to be equivalent. Therefore, we can prove this equivalence by a mere renaming of variables. Renaming p to q and q to p, we obtain:

$$p \implies q \equiv \neg q \implies \neg p.$$

Therefore, we have found that a conditional is equivalent to its contrapositive. To make sense of this, consider the rain example. If the grass doesn't get wet, then, indeed, it must not have rained: for if it *had* rained, the grass would have been wet.

Recall the rule of modus tollens: if we know that $p \implies q$ and also that q is false, we can deduce that p is false. Now we see clearly why this rule works: to say that $p \implies q$ is to say that $\neg q \implies \neg p$, so if q is false (i.e., $\neg q$ is true), then $\neg p$ follows by modus ponens.

The equivalence between a conditional and its contrapositive is very useful for proving conditional claims. When we want to show a conditional of the form $A \implies B$, the most natural way to proceed is by what is called *direct proof*: we assume that A is true, and see if we can deduce B. If we can, we conclude that $A \implies B$. But sometimes, it is easier to prove the contrapositive: to assume that B is false and thereby deduce that A is false. If we can do so, we conclude $\neg B \implies \neg A$, and therefore conclude that $A \implies B$. This is called *proof by contrapositive*, and is one example of an *indirect proof*.

6.27 This equivalence states that to deny a conditional is to claim that its premise is true and its conclusion false. Given our previous experience with the conditional, we should expect this to be accurate. We will prove it deductively:

$$\begin{aligned} \neg(p \implies q) &\equiv \neg(\neg(p \wedge \neg q)) && \text{by definition.} \\ &\equiv p \wedge \neg q && \text{by double denial.} \end{aligned}$$

That is, to claim a conditional is to claim that it *isn't* the case that its premise is true and its conclusion is false, so to deny a conditional is to say that this *is* the case.

6.28 Here we have an alternate characterization of the denial of a conditional, which claims that it is equivalent to that conditional's inverse. Let's rewrite this proposed equivalence in terms of conjunction: perhaps various denials will cancel out and things will

become clearer.

$$\begin{aligned}
\neg(p \implies q) &\equiv \neg p \implies \neg q && \\
p \wedge \neg q &\equiv \neg p \implies \neg q && \text{by the previous equivalence.} \\
p \wedge \neg q &\equiv \neg(\neg p \wedge \neg\neg q) && \text{by definition.} \\
p \wedge \neg q &\equiv \neg\neg p \vee \neg\neg\neg q && \text{by De Morgan's first law.} \\
p \wedge \neg q &\equiv p \vee \neg q && \text{by double denial, twice.}
\end{aligned}$$

But if this were true, conjunction and disjunction would be the same, which we know is not true. Therefore, our original assumption at the beginning of these deductions, namely that $\neg(p \implies q) \equiv \neg p \implies \neg q$, must be false.*

To understand this intuitively, notice that the right-hand formula of the original equivalence, $\neg p \implies \neg q$, is (contrapositively) equivalent to $q \implies p$. But it cannot be true that $\neg(p \implies q) \equiv q \implies p$: for if it were, biconditionals would be impossible. That is, if we were to learn that $q \implies p$ was true, then we would be forced to conclude that $p \implies q$ was false, and this would rule out the possibility of the biconditional.

If it is not immediately clear why that would be a problem, consider the case where the premise and conclusion are the same. If we were to learn that $p \implies p$ was true (which indeed it always is), we would be forced to conclude that the converse, $p \implies p$, was false.†

6.29 Intuitively,‡ a proposition of the form $(p \vee q) \implies r$ claims that, if *either* of p and q are true, then r must follow.

A proposition of the form $(p \implies r) \wedge (q \implies r)$ claims that p implies r and also q implies r.

Let's consult a truth table.§

p	q	r	$(p \vee q) \implies r$	$(p \implies r) \wedge (q \implies r)$
$\bot$	$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\bot$	$\top$	$\top$	$\top$
$\bot$	$\top$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\bot$	$\bot$
$\top$	$\bot$	$\top$	$\top$	$\top$
$\top$	$\top$	$\bot$	$\bot$	$\bot$
$\top$	$\top$	$\top$	$\top$	$\top$

Indeed, it is true.

One might ask, if it were true that $(p \vee q) \implies r$, how would this fact be enforced by reality? r must follow whenever *either* p or q is true (or both), regardless of which one. The only way to enforce this is if $p \implies r$ and also $q \implies r$: that way, if either premise is true, r will follow from the corresponding conditional.

*

Here we are once again employing proof by contradiction.

†

Due to its symmetry, $p \implies p$ is its own converse.

‡

And again pretending that a conditional is an English *if... then* statement for the sake of intuition.

§

We could use deduction instead, but this is one case where it is easier to just make the table.

6.30 Our final equivalence states that to claim the biconditional between two propositions is to claim that they are either both true or both false.

This is easiest to show with a truth table.

p	q	$p \iff q$	$(p \land q) \lor (\neg p \land \neg q)$
$\bot$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\bot$	$\bot$
$\top$	$\bot$	$\bot$	$\bot$
$\top$	$\top$	$\top$	$\top$

We see that it is true, and this should come as no surprise, because we have long since established that a biconditional claims that its operands have the same truth value: therefore they must either be both true or both false.

6.31 They are closely related: indeed, when I was first learning logic I didn't see the need for the '$\equiv$' symbol at all. Both claim that two things have the same truth value.

However, there is an important difference. When you write a biconditional between two formulas, the result is another formula, and is therefore neither true nor false.* Rather, it may be true for some combinations of inputs (when those inputs cause the two sides of the biconditional to have the same truth value) and false for others. But when you claim that two formulas are equivalent, the result is a proposition: either they are equivalent (they *always* have the same truth value) or they are not.

In other words, a logical equivalence has to look at all possible inputs at once to determine its single truth value, while a biconditional has more nuance: it can have different truth values for each combination of inputs. Very soon, we will clarify this relationship further.

7.3 A contradictory formula is one that is always false: that is, whose truth table column has $\bot$ in every row.†

7.5 [`your formula`] $\equiv \bot$.

7.7 This equivalence states that, if you claim that two propositions are true, but one of them is false, your claim is false regardless of the truth value of the other proposition. Naturally, this is correct.

7.8 This equivalence states that, if you claim that two propositions are true, and one of them is, then your claim is true regardless of the truth value of the other proposition. This is incorrect, because if the other proposition isn't true, the conjunction is false.

7.9 This equivalence states that, if you claim that two propositions are true, and one of them is, then the truth value of your claim is the truth value of the other proposition.

Consider the truth table.

p	$p \land \top$
$\bot$	$\bot$
$\top$	$\top$

*

Unless it is tautological or contradictory.

†

Tautologies and contradictions are the only exceptions to the rule that truth values are determined by reality, because they are always true or always false (respectively) regardless.

It is true.

This makes sense, because if the other proposition is true, then both are true and hence their conjunction is true. If the other proposition is false, then the conjunction is immediately false.

7.10 This equivalence states that, if you claim that at least one of two propositions is true, and one of them *is*, then your claim is true regardless of the truth value of the other proposition. This is correct: as soon as one operand is true, the disjunction is immediately true.

7.11 This equivalence states that, if you claim that at least one of two propositions is true, and one of them *is*, then the truth value of your claim is the truth value of the other proposition. As we saw in the previous exercise, this claim is incorrect. If one operand is true, then the disjunction will be true even if the other operand is false.

7.12 This equivalence states that, if you claim that at least one of two propositions is true, and one of them is false, your claim is false regardless of the truth value of the other proposition. This is incorrect: if the other proposition is true, then the disjunction will be satisfied even though one proposition is false.

7.13 This equivalence states that, if you claim that at least one of two propositions is true, and one of them is false, then the truth value of your claim is the truth value of the other proposition.

This is correct. If the other proposition is true, then the disjunction is immediately true. If the other proposition is false, then neither proposition is true and the disjunction is falsified.

7.14 This equivalence states that, if you claim that p is true and also that one of p or q must be true, then this is the same as just claiming p.

Here is the truth table.

p	q	$p \wedge (p \vee q)$
$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\bot$
$\top$	$\bot$	$\top$
$\top$	$\top$	$\top$

Therefore, it is true.

This is fairly intuitive. In any circumstance where $p \wedge (p \vee q)$ is true, p must also be true (since otherwise $p \wedge \ldots$ wouldn't be satisfied). And in any circumstance where p is true, $p \vee q$ is satisfied so $p \wedge (p \vee q)$ must also be satisfied.

7.15 This equivalence states that, if you claim that either p is true or *both* p and q are true, then this is the same as just claiming p. Let us investigate this deductively.

$$\begin{aligned} p \vee (p \wedge q) &\equiv (p \vee p) \wedge (p \vee q) && \text{by the distributivity of disjunction over conjunction.} \\ &\equiv p \wedge (p \vee q) && \text{by the redundancy of } p \vee p. \\ &\equiv p && \text{as found in the previous exercise.} \end{aligned}$$

Therefore, indeed, $p \vee (p \wedge q) \equiv p$.

This makes sense, because in any circumstance where $p \vee (p \wedge q)$ is true, it must be that either p is true or $p \wedge q$ is true: in either case, p is true. And in any circumstance where p is true, $p \vee (p \wedge q)$ is satisfied immediately.

7.16 This equivalence states that $p \wedge \neg p$ is contradictory: that is, that no statement can be both true and false. Hence, it is a formal statement of the Law of Noncontradiction. We would therefore expect it to be true, and indeed:

p	$\neg p$	$p \wedge \neg p$
$\bot$	$\top$	$\bot$
$\top$	$\bot$	$\bot$

We find that $p \wedge \neg p$ is always false, and is therefore contradictory by definition.

This makes sense: if p is false, then $p \wedge \neg p$ is immediately falsified, and if p is true, then $\neg p$ is false and $p \wedge \neg p$ is once again falsified.*

Alternatively, we could have proven this deductively:

$$\begin{aligned} \neg(p \wedge \neg p) &\equiv \neg p \vee \neg\neg p && \text{by De Morgan's first law.} \\ &\equiv \neg p \vee p && \text{by double denial.} \\ &\equiv p \vee \neg p && \text{by the commutativity of disjunction.} \\ &\equiv \top && \text{by tautology.} \end{aligned}$$

That is, the *denial* of $p \wedge \neg p$ is a tautology: therefore, it is itself a contradiction.

7.17 This equivalence states that $p \implies p$ is tautological: that is, that every proposition implies itself. Let's rewrite this using some familiar equivalences:

$$p \implies p \equiv \neg(p \wedge \neg p) \qquad \text{by definition.}$$

It is exactly the denial of the contradiction we just found! Furthermore, as we found in the previous exercise, this denial is equivalent to the first tautology we discovered, $p \vee \neg p$.† Therefore, it is tautological, so the equivalence proposed in this exercise is true.

7.18 This equivalence states that $p \implies \neg p$ is a contradiction: that is, that no proposition implies its denial. Given our previous success with rewriting contradictions and tautologies and finding them equivalent to previously discovered ones, we will take the

*

Here I am essentially performing a proof by cases.

†

In mathematics, everything is connected.

same approach here.

$$\begin{aligned} p \implies \neg p &\equiv \neg(p \wedge \neg\neg p) && \text{by definition.} \\ &\equiv \neg(p \wedge p) && \text{by double denial.} \\ &\equiv \neg p && \text{by the redundancy of } p \wedge p. \end{aligned}$$

Perhaps surprisingly, given our previous momentum, this proposed contradiction is incorrect. Rather, $p \implies \neg p \equiv \neg p$: to say that a proposition implies its denial is not necessarily wrong, but is simply to deny p.

This is one of the strange results of the difference between the conditional and the English *if... then* statement. If p is false, then $p \implies \ldots$ is true regardless of the conclusion, as we have found before. In such a circumstance, $p \implies \neg p$ is true, and so is $\neg p$. If, on the other hand, p is true, then $p \implies \neg p$ becomes $\top \implies \bot$, which is of course false, as is $\neg p$. So in either case, the equivalence holds.

7.19 This equivalence states that to say that p implies a falsehood is to deny p.

$$\begin{aligned} p \implies \bot &\equiv \neg(p \wedge \neg\bot) && \text{by definition.} \\ &\equiv \neg(p \wedge \top) \\ &\equiv \neg p && \text{since } p \wedge \top \equiv p. \end{aligned}$$

So it is true, and indeed it is one justification of the rule of modus tollens: to say that p implies a falsehood is to say that p is false.

7.20 This equivalence states that to say that p implies a truth is to affirm p.

$$\begin{aligned} p \implies \top &\equiv \neg(p \wedge \neg\top) \\ &\equiv \neg(p \wedge \bot) \\ &\equiv \neg\bot && \text{because } p \wedge \bot \text{ is always false.} \\ &\equiv \top. \end{aligned}$$

Instead of p, we have found $\top$: an unexpected tautology. Therefore, the proposed equivalence is false.

This tautology we found is yet another artifact of the difference between the conditional and the English *if... then* statement. As we found previously, if the conclusion of a conditional is true, then that conditional must itself be true: the only way a conditional can be false is if the premise is true *and the conclusion is false*. Of course, in English we demand some relation between the premise and conclusion, not merely that the conclusion is true.

7.21 This equivalence claims that $\bot \implies p$ is tautological: that is, that a falsehood implies everything.

$$\begin{aligned} \bot \implies p &\equiv \neg(\bot \wedge \neg p) \\ &\equiv \neg\bot \\ &\equiv \top. \end{aligned}$$

It is true, and this fact is known as the *Principle of Explosion.* For if we were to prove some proposition both true and false, then:

1. By its falsehood, it would *imply* every conceivable proposition.
2. By its truth, all those propositions would therefore be true (according to modus ponens). This simultaneous proof of all possible propositions, true, false, necessary or impossible, is what we mean by *explosion.*

7.22 This equivalence claims that $\top \implies p$ is tautological: that is, that a truth implies everything. This is obviously incorrect: if it were true, then everything would be true, and this is not the case.

To verify our intuition, let's check the truth table.

p	$\top \implies p$
$\bot$	$\bot$
$\top$	$\top$

Rather, we find that $\top \implies p \equiv p$: to say that the truth implies p is to say that p is true. This makes sense, given the rule of modus ponens.

7.23 This equivalence claims that to say that p is both true and false is to say that p implies its denial. We have previously found $p \wedge \neg p$ to be a contradiction, and found $p \implies \neg p \equiv p$. Since the latter is not a contradiction, the proposed equivalence is false. If we want, we can easily verify this with a truth table:

p	q	$p \wedge \neg p$	$p \implies \neg p$
$\bot$	$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\bot$	$\top$
$\top$	$\bot$	$\bot$	$\bot$
$\top$	$\top$	$\bot$	$\bot$

As expected, the columns differ.

8.1 If p is true, that doesn't necessarily mean q is true. So we wouldn't expect this to be true. To verify this intuition, we would make a truth table for $p \implies q$ and see if it's tautological. But we know very well that it isn't, so we won't bother.

8.2 We have already found in Exercise 7.17 that $p \implies p \equiv \top$, which is to say that $p \implies p$ is tautological. Therefore, p does logically imply p, i.e., $p \models p$.

8.3 This is the first exercise really demanding that you understand the difference between logical equivalence and implication. To see this, let us construct the truth table for $p \implies (p \vee q)$.

p	q	$p \vee q$	$p \implies (p \vee q)$
$\bot$	$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\top$	$\top$
$\top$	$\top$	$\top$	$\top$

As you can see, p and $p \vee q$ are not logically *equivalent*, because their columns differ. However, it is important to notice that the differences only occur in rows where p is false: when p is true, $p \vee q$ is also true.* Having made this observation, we can immediately conclude that $p \models p \vee q$.† And indeed, the final column shows that $p \implies (p \vee q)$ is tautological.

8.4 If $p \vee q$ is true, does that mean p is true? No: it could be that q is true (so that $p \vee q$ is true) while p is false. That is, we can imagine a circumstance in which $(p \vee q) \implies p$ is false, and therefore we expect this proposed logical implication to be incorrect.

Let us construct the truth table.

p	q	$p \vee q$	$(p \vee q) \implies p$
$\bot$	$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\top$	$\bot$
$\top$	$\bot$	$\top$	$\top$
$\top$	$\top$	$\top$	$\top$

We see that indeed, the conditional is true in all cases except when q is true and p is false. The conditional is not tautological, so we conclude that the proposed implication is false.‡

8.5 This implication is a formal justification of the rule of modus ponens. To verify it, we construct a truth table for the formula $[p \wedge (p \implies q)] \implies q$, and check if it is tautological.

p	q	$[p \wedge (p \implies q)] \implies q$
$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\top$
$\top$	$\bot$	$\top$
$\top$	$\top$	$\top$

Thankfully, it is.

8.6 Imagine that if p is true then q is true, and that if q is true then r is true. Then, if p were true, we would know that q was true, and therefore that r was true. Therefore, this proposed logical implication makes sense, intuitively.

To verify it, we make the truth table.

*

This makes sense, because if p is true, we have satisfied one of the two options required by $p \vee q$.

†

Make sure you think about why.

‡

Notice that this exercise is similar to the previous one: the premise and conclusion have merely swapped places. This highlights another difference between logical equivalence and implication: logical equivalence is symmetrical, whereas logical implication is not. That is, if $A \equiv B$ then necessarily $B \equiv A$, whereas it is possible that $A \models B$ without $B \models A$. Intuitively, this is because logical equivalence happens when two formulas have *the same* column in their truth table, and having something *in common* is symmetric. But logical *implication* is a relation between a *premise* and a *conclusion* such that whenever the premise is true, the conclusion is also always true. The difference in how premise and conclusion are treated is what makes logical implication non-symmetric.

p	q	r	$p \implies q$	$q \implies r$	$p \implies r$	$[(p \implies q) \wedge (q \implies r)] \implies (p \implies r)$
$\bot$	$\bot$	$\bot$	$\top$	$\top$	$\top$	$\top$
$\bot$	$\bot$	$\top$	$\top$	$\top$	$\top$	$\top$
$\bot$	$\top$	$\bot$	$\top$	$\bot$	$\top$	$\top$
$\bot$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\bot$	$\top$	$\bot$	$\top$
$\top$	$\bot$	$\top$	$\bot$	$\top$	$\top$	$\top$
$\top$	$\top$	$\bot$	$\top$	$\bot$	$\bot$	$\top$
$\top$	$\top$	$\top$	$\top$	$\top$	$\top$	$\top$

*

Because we aren't operating upon it with any logical operators.

10.1

1. $\forall$ is an upside-down letter A, which stands for the word "all."
2. $\exists$ is a backwards letter E, which stands for the word "exists."

The letters are flipped to avoid confusion with variables named by the same letters.

10.2

1. $(\exists x)(x$ is a unicorn$)$.
2. $(\exists x)(x$ is red$)$.
3. $(\forall x)(x$ is red$)$.

10.3

1. I suspect this is false because, as far as I know, there does not exist even a single unicorn. However, I cannot be certain.
2. This is definitely true, because my hair is red. That is, I have found an example of, also known as a *witness* to, the existential claim, which is enough to verify it.
3. This is definitely false, because my filing cabinet is black, not red. That is, I have found a *counterexample* to the universal claim, which is enough to falsify it.

11.1 The answer is that it produces a new predicate that is *composite*.

As discussed previously, when we apply a composite predicate to a subject, we produce a composite proposition whose truth value is determined by:

1. Determining the truth value of the application of the original (less-composite) predicate.
2. Operating upon this truth value with the logical operators according to the structure of the composite predicate.

For example, if we take the predicate "x is red," and apply it to "the firetruck," the resulting predicate is (presumably) true. Therefore, when we apply the denial of that same predicate to that same subject the resulting proposition will be false. Other operators behave correspondingly.

But if we quantify a composite predicate, rather than directly applying it, the result is not a composite proposition* but an atomic proposition. Recall that I said one of the reasons we study propositional logic (in which we learn about logical operators) before predicate logic (in which we learn about atomic propositions) is that logical operators are necessary for constructing certain atomic propositions. This is why: we can use logical operators to construct composite predicates, and quantify these to produce atomic propositions.

11.2

1. This is true. For example, if all objects were red, and all objects were heavy, then it would follow that every object is both red and heavy. Conversely, if every object were both red and heavy, it would follow that every object is red and also that every object is heavy.

2. This is false. Take for example $P(x) \iff$ "x is red" and $Q(x) \iff$ "x is blue." Then the left-hand side of the biconditional claims that either every single object is red, or that they are all blue, or that every single object is both blue and red. That is, all the objects have to choose *together* what their color will be as a group. They cannot choose individually, with one object being blue and another being red. But the right-hand side claims something different: that every individual object must be red or blue (or both), but not neither.

   The left hand side does imply the right: if the objects all make a choice together, they still made choices individually. But the right does not imply the left, because the right allows a mixture and the left does not.

3. False. The left-hand side says that some element satisfies P and some element satisfies Q, but not that they are the same element.

   The right-hand side makes a stronger claim, that there is a single element that simultaneously satisfies both P and Q.

   If the right is true, the left is true. But the left can be true without the right.

4. True. The left side says that either there's some object satisfying P or some object satisfying Q (or both). But if there is such an object, then it would also satisfy $P \vee Q$. Conversely, if some object satisfies $P \vee Q$, as the right side claims, then that object must either satisfy P or satisfy Q (or both). Otherwise, it could not satisfy $P \vee Q$.

5. False. If P applies to everything and Q applies to some x, then P must also apply to that x. Therefore, the left side does imply the right side. However, the converse is obviously not true.

11.4

1. This is false, of course. Yet there is a kernel of truth. If we assume, or know, that there is any object α that exists, then by the left-hand proposition we know that $P(\alpha)$. Therefore $(\exists x)P(x)$: namely, this object α.

2. False. The forward direction does hold: if P always implies Q, then if P is always true, Q must also always be true.

   But, if P always being true makes Q always true, that doesn't imply that, for every individual object, P implies Q. It is possible that there is something special about P being true for *all* objects that would cause Q to then be true for all objects, but that P is *not* always true and there is an object that satisfies P but not Q.

3. False. The forward direction holds. If P and Q always have the same truth value, then the claims that "P is always true" and that "Q is always true" must also have the same truth value.

   However, it might be that the right hand side is true just because neither P nor Q are always true, and that they are true for different objects.

4. False. Neither direction holds. It is possible that there is some single object that, if it satisfied P, would also satisfy Q, but that this object does *not* satisfy P, and that in fact there is some other object that satisfies P, while there is no object that satisfies Q.

   It is also possible that, if any object were to satisfy P then there would also have to be an object satisfying Q, but that these could be different objects and that there is no single object which, if it satisfied P would also have to satisfy Q.

11.5 Not important at all: the names of variables don't matter, as long as they are named consistently. By "consistently" I mean that the same object should not be given multiple names, nor should multiple objects be given the same name. For instance, in the first part of Exercise 11.2, in $(\forall x)(P(x) \wedge Q(x))$, only one object is being spoken of: x. The three instances of the variable x all refer to that same object, and the predicates P and Q are each asserted of it. In such a circumstance, it's fine for us to rename *all three* usages of x to y (or any other label), but we cannot rename only one or two, or rename each to a different label, without changing the meaning of the proposition. By giving them two different names, we would be implying that the objects are, or at least *could be*, distinct.

12.1 This is false. The left says that every x has some corresponding object (y) such that P holds. That is, there is a sort of pairing between each x and another object in order to satisfy P. The right says that there is some y such that P holds for every single other object (x). That is, y doesn't have some special object it must pair with to satisfy P: any one will do.

The left does not imply the right. If every x has a y to satisfy P with, that doesn't mean y could satisfy P with any object: it might need this x specifically. However, the right *does* imply the left. If there is some y that satisfies P along with *any* object, then for any given x, it can find an object to pair with: namely, y.

12.2 This is true. The two propositions on either side of the biconditional are logically identical: the only difference is the particular labels of the variables, and that has no logical relevance.

It is important to notice the scope of the quantifications here. The two left-hand quantifications and the two right-hand quantifications are in separate parenthesis-groups. Therefore, there is no connection between the instance of x on the left and the instance on the right. That is, there is no more of a (logical) relationship between the two instances of x than between the left-hand instance of x and the right-hand instance of y.

Chapter 2: Introduction to set theory

6.1 A self-denying proposition is false whenever it is true, and true whenever it is false: therefore, it must be either both true and false, or neither. These options are generally considered impossible, but even if we open our minds enough to allow them, we are not saved from paradox.

*

This is a reiteration of the principle of explosion explained in the last chapter.

If it is neither true nor false, then it is certainly not *false*. But then, because the proposition claims itself to be false, that claim would be inaccurate: in other words, it would be false, and therefore true, and therefore both. Therefore, a self-denying proposition must be both true and false.

But if some proposition P is false, then the conditional $P \implies Q$ is true for every proposition Q. And if P is *also true*, then we must conclude Q by modus ponens. Therefore, the mere expressibility of a self-denying proposition results in the simultaneous proof of all propositions, regardless of their correspondence to reality.*

Therefore, within these original set theories one could prove not only every mathematical truth, but absolutely every proposition. This defeats the purpose of a mathematical theory, which is to only allow one to prove *true* theorems, so that by their derivation we can learn about the intuitive ideas to which they indirectly refer. Having proven some apparent fact, we would have to wonder: is it provable because it's true, or because our theory is broken?

Yet actually, this is not a serious problem! Modern mathematicians, scientists and philosophers have developed a powerful multidisciplinary technique called *proof by intuition*. In our context, here is how it works:

1. Use the axiom schema of comprehension.
2. If your results make sense, they are true.
3. If your results don't make sense, your chosen predicate must have been bad. Don't choose that predicate.

But actually, we began this discussion with a significant mistake. Mathematicians tend to think that every proposition must be either true, false, both, or neither—even if you disagree with the "laws" of logic. To see this, let's make a simple truth table:

P is true	**P** is false	Status of **P**
$\bot$	$\bot$	**P** is neither true nor false.
$\bot$	$\top$	**P** is false.
$\top$	$\bot$	**P** is true.
$\top$	$\top$	**P** is both true and false.

But this is circular! The truth table is constructed on the hidden assumption that the two claims "**P** is true" and "**P** is false" must each be either true or false: the combinations of these possibilities correspond to the table's rows. And this assumption is the very thing sought to be proved by the table.

Even if we get rid of the table, people are still generally left with an *intuition* that surely these are the only possibilities. But this is the same intuition that goes further and claims that every proposition must be either true or false but not both: the very intuition undermined by the Liar's Paradox. If we are to successfully resolve that paradox once and for all, we will require a radical change of intuition.

Chapter 3: Extensionality

7.1 blah

List of incorrect solutions

Chapter 1: Mathematics and logic

6.20.

Chapter 2: Introduction to set theory

6.1.

Chapter 3: Extensionality